

**GANDHI INSTITUTE OF ENGINEERING
AND TECHNOLOGY UNIVERSITY,
ODISHA, GUNUPUR**

DEPARTMENT

OF

**COMPUTER SCIENCE &
ENGINEERING**

(SCHOOL OF ENGINEERING & TECHNOLOGY)



BACHELOR'S DEGREE PROGRAMME

(B. Tech- CSE- Data Science)

**Course Structure and Syllabus
Academic Session**

[2023-27]



GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY, ODISHA, GUNUPUR

SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

1. Institute Vision & Mission, Department Vision & Mission, PEO's, PO's, PSO's

Institute Vision:

To be a renowned and Globally Recognized University giving importance to academic excellence with latest technology, Research, Innovation and Entrepreneurial Attitude.

Institute Mission:

- To create innovative and committed work force to cater to the societal, environmental and economic needs of the Nation.
- To promote Education and Research Globally at par with the international standards
- To prepare the future leaders with latest skills befitting to become Entrepreneurs or Employable.
- To support and uplift the meritorious students of this Tribal Area to represent as an Ambassador in all forums as a part of our social responsibility.

Department Vision:

To be globally acclaimed as a premier Computer Science and Engineering department through excellent teaching and strong research environment by creating professionally competent engineers with moral values committed to build a vibrant nation.

Department Mission:

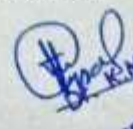
- Provide a positive and professional learning environment to master the fundamental principle of computing, problem solving skills, leadership qualities, team-spirit, ethics and societal responsibilities.
- To enable the graduates to use modern tools and technologies in the area of computer science and information technology.
- Establish Industry Institute Interaction program to enhance the employability skills.
- To promote innovative research in Computer Science and Engineering to serve the needs of Industry, Government and Society at large.

Program Educational Objectives:

- To render best platform for adequate training and opportunities to work as teams on projects with effective communication skills and leadership qualities and understand professional ethics, social awareness and organizational context in which their engineering skills are utilized.

To endow the students with sound knowledge in the field of mathematics, basic science and engineering fundamentals to solve and inculcate the ability to utilize their skills to prepare them for higher studies, research and analyze engineering problems.




K. MURALI GOPAL
MOD. CSE
GIET UNIVERSITY
GUNUPUR-765022

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

- To extend an ability to analyze the need of the society by providing innovation solutions, leading to their personal cum professional growth as an entrepreneur.

Program Outcome:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

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- 11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- 12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes:

- To provide students an understanding of the expectations of industry and practical competence with a broad range of programming language and open source platforms through value added courses.
- The ability to analyze and develop computer programs in the areas related to Artificial Intelligence, Big data Analysis and Cyber Security for efficient design of computer-based systems of varying complexity.

Program Educational Objectives:

- Develop proficiency in the field of mathematics, basic science, and engineering fundamentals to analyze and solve a wide range of computational problems in industry or other work environments and succeed in an advanced degree program in a field such as engineering.
- Possess the ability to collaborate as team members and team leaders for facilitating technical solutions for computing systems with awareness and commitment to their ethical responsibilities.
- Attain the ability to critically analyze the need of society and provide innovative solutions in multi-disciplinary areas adopting technologies leading to their personal and professional growth as an entrepreneur.

III Semester [Second Year]
B. Tech in Computer Science and Engineering (Data Science) (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	BS	23BBSBS23003	Applied Statistics	3	1	0	4	40	60
2	PC	23BCDPC23001	Data Base Management Systems	3	0	0	3	40	60
3	ES	23BECES23001	Digital Electronics	3	0	0	3	40	60
4	PC	23BCDPC23002	Object Oriented Programming through Java	3	0	0	3	40	60
5	PE	23BCDPC23011/23BCDPC23012	PE-I (From Bucket)	3	0	0	3	40	60
6	HS	23BBSHS23001	Intermediate communication skills and critical thinking	1	0	0	1		
			TOTAL	16	1	0	17	200	300
PRACTICAL/SESSIONAL									
7	ES	23BECES23001	Digital Electronics Lab	0	0	2	1	30	20
8	PC	23BCDPC23001	Data Base Management Systems Lab	0	0	2	1	30	20
9	PC	23BCDPC23002	Object Oriented Programming through Java Lab	0	0	2	1	30	20
10	HS	23BBSHS23001	Intermediate communication skills and critical thinking	0	0	2	1	30	20
11	PST	23BCHPW23101	Summer Internship-I	0	0	0	1	30	20
12	PST	23BCSLP23101	Learning Project-I	0	0	0	1	50	0
			TOTAL	0	0	8	6	200	100
			TOTAL	16	1	8	23	400	400

Professional Elective Bucket

Sl.	Sem	Professional Elective I
1	23BCDPC23011	Introduction to Data Science
2	23BCDPC23012	Data Preparation and Analysis

Subject Code	Object Oriented Programming Through Java	L	T	P	C									
		3	0	2	4									
Course Educational Objectives														
CEO 01	The model of object oriented programming: abstract data types, encapsulation, inheritance and polymorphism													
CEO 02	Fundamental features of an object oriented language like Java: object classes and interfaces, exceptions and libraries of object collections													
CEO 03	How to take the statement of a business problem and from this determine suitable logic for solving the problem; then be able to proceed to code that logic as a program written in Java.													
CEO 04														
Course Outcomes: Upon successful completion of this course, students should be able to:														
CO 01	Knowledge of the structure and model of the Java programming language.													
CO 02	Use the Java programming language for various programming technologies.													
CO 03	Develop software in the Java programming language.													
CO 04	Evaluate user requirements for software functionality required to decide whether the Java programming language can meet user requirements.													
CO 05	Propose the use of certain technologies by implementing them in the Java programming language to solve the given problem.													
CO 06	Choose an engineering approach to solving problems, starting from the acquired knowledge of programming.													
CO-PO & PSO Mapping (Articulation Matrix):														
COs	PROGRAMME OUTCOMES												PSOs	
	01	02	03	04	05	06	07	08	09	10	11	12	01	02
01	3	1											1	
02	3	2											1	
03	2	2	2										2	
04	2	1	2										1	
05	2	1	3										1	
06	1	3	3										1	
Avg.	2.1	1.6	2.5										1.1	
Syllabus														
UNIT I												13 Hours		
An introduction to Object Oriented Programming, Features of Object Oriented Programming Introduction to Java. Difference between C/C++ and Java, Features of Java, First Java Program, Writing the java program, Compiling the program, JVM and its significance in executing a program, Architecture of JVM. Taking Input from keyboard, Command Line Arguments, Using Scanner Class, Using Buffered Reader class. Java Tokens, Datatypes, Operators, Control Structures, Conditional Statements, Iteration Statement/Loops, Jumping Statements, Java Arrays, 1-dimensional arrays, 2-dimensioanl arrays.														

UNIT II		15 Hours
Introduction to Classes and Objects. Constructors, static Keyword, this Keyword, Array of Objects, Access Modifiers (Public, Private, Protected, Default). Inheritance, Types of Inheritance and Java-supported Inheritance, super, Polymorphism, Method Overloading, Constructor Overloading, Method Overriding, Dynamic Method Dispatching. Interfaces, Multiple Inheritance Using Interfaces, abstract classes. String Handling in Java: String, StringBuffer, StringBuilder, StringTokenizer Wrapper classes, Auto boxing and unboxing.		
UNIT III		12 Hours
Java API Packages, User-Defined Packages, Accessing Packages. Error and Exception Handling, Types of exceptions Hierarchy of Exception classes, try, catch, finally, throw, throws, Commonly used Exceptions and their details ,User defined exception classes. Multithreading, Thread in Java, Thread execution prevention methods. (yield(), join(), sleep()), Concept of Synchronization, Inter Thread Communication, Basics of Deadlock, Demon Thread. Inner Classes: Introduction, Member inner class, Static inner class, Local inner class, Anonymous inner class. IO Streams (java.io package), Byte Stream and Character Stream, Files and Random Access Files, Serialization.		
UNIT IV		12 Hours
Collection Frame Work (java.util), Util Package interfaces, List, Set, Map. Applet Introduction, Life Cycle of an Applet, GUI with an Applet. AWT : The AWT class hierarchy, user interface components- Labels, Button, Text Components, Check Box, Check Box Group, Choice, List Box, Panels – Scroll Pane, Menu, Scroll Bar. Working with Frame class, Colour, Fonts and layout managers. Event Handling: Events, Event sources, Event Listeners, Event Delegation Model (EDM), Handling Mouse and Keyboard Events, Adapter classes		

Subject Code		Name of the Subject						L	T	P	C				
		Database Management System						3	0	1	4				
Course Educational Objectives															
Pre -Requisite: A student should have basic idea on programming and file systems.															
CEO 1	Gain a good understanding of the architecture and functioning of Database Management Systems as well as associated tools and techniques.														
CEO 2	Understand and apply the principles of data modeling using Entity Relationship and develop a good database design.														
CEO3	Understand the use of Structured Query Language (SQL) and its syntax														
CEO4	Apply Normalization techniques for effective database design.														
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Identify and Classify the concepts of Database Management system, Data models and architecture of database, ER to Relational mapping concepts.														
CO2	Applying The constraints in database using different query languages like:- relational algebra and calculus implementing the Data definition and data manipulate languages in Database.														
CO3	Compare the different normal forms to Apply normalization process to construct consistent Database.														
CO4	Design and Develop the Database with concurrency control and recovery strategies to make complete Database without conflict and anomalies in concurrent access environment.														
CO5	Develop efficient storage scheme of saving and retrieving Records and Files														
CO6	Apply the principles of database transaction management, database recovery and security.														
CO-PO & PSO Mapping:															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	1	1												
CO2	3	2	1												
CO3	2	1	1												
CO4	2	1	2												
CO5	2	1	2												
CO6	3	2	1												
Avg.	2.5	1.33	1.33												
SYLLABUS															

UNIT:1 Introduction to database Systems: advantages of database system over traditional file system, Basic concepts & Definitions, Database users, Database Language, Database System Architecture, Schemas, Sub Schemas, & Instances, database constraints, 3-level database architecture, Data Abstraction, Data Independence, Mappings, Structure, Components & functions of DBMS, Data models. Entity-Relationship model: Basic concepts, Design process, Constraints, Keys, Design issues, E-R Diagrams, weak entity sets, Extended E-R features – generalization, specialization, aggregation, Reduction to E-R database schema.	(6 Hours)
UNIT:2 Relational Query Languages: Relational Operations. Relational Algebra, Selection and projection set operations, renaming, Joins, Division, Examples of Algebra overviews, Relational calculus, Tuple relational Calculus, Domain relational calculus, Extended relation algebra operation	(10 Hours)
UNIT:3 Normalization: Introduction, non-loss decomposition and functional dependencies, First, Second, and third normal forms – dependency preservation, Boyce-Codd normal form. Higher Normal Forms - Introduction, Multi-valued dependencies and fourth normal form, Join dependencies and fifth normal form	(10 Hours)
UNIT:4 Transaction Concept: Transaction properties (ACID), Transaction State, Schedules, Serializability (Conflict Serializability and View Serializability), Need for Concurrency, Concurrency Control- Lock based control, time stamping Protocol, Two Phase Locking, Dead lock in DBMS Recovery and Atomicity: Log Based Recovery, Transaction Logs and Shadow Paging, Types of Recovery Techniques in DBMS, Rollback/Undo Recovery Technique, Commit/Redo Recovery Technique, Checkpoint Recovery, Backup Techniques	(10 Hours)
UNIT:5 Detailed Storage Architecture: Storage Strategies: Detailed Storage Architecture, Storing Data, Magnetic Disk, RAID, Other Disks, Magnetic Tape, Storage Access, File & Record Organization, File Organizations & Indexes, Order Indices, B+ Tree Index Files, Hashing Data Dictionary	(6 Hours)
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Sudarshan, Korth: Database System Concepts, 6th edition, McGraw-Hill Education 2. Elmasari&Navathe: Fundamentals of Database System, Pearson Education.	
Reference Books: 1. Elmasari&Navathe: Fundamentals of Database System, Pearson Education. 2. Ramakrishnan: Database Management Systems, McGraw-Hill Education. 3. Andrew S. Tanenbaum: Modern Operating Systems, 3rd Edition, Pearson Education. 4. Terry Dawson, Olaf Kirch: Linux Network Administrator's Guide, 3rd Edition O'Reilly	

Subject Code		SUBJECT NAME											L	T	P	C
		Introduction to Data Science												4		4
Course Educational Objectives																
CEO 01	Able to apply fundamental algorithmic ideas to process data															
CEO 02	Able to construct predictive models to classify new data set															
CEO 03	Learn to apply hypotheses and data into actionable predictions															
CEO 04	Document and communicate the results effectively to different stakeholders and effectively communicate the findings using visualization techniques.															
Course Outcomes: : Upon successful completion of this course, students should be able to:																
CO1	Understand the Foundations of Data Science. Conduct Data Acquisition and Pre-processing															
CO2	Describe the data using various statistical measures															
CO3	Work Collaboratively in Data Projects. Use Relevant Data Science Tools and Technologies															
CO4	Apply Data Modelling Techniques. Evaluate and Select the appropriate Model.															
CO5	Perform data reduction and apply visualization techniques															
CO6	Apply Data Science Skills to Real-World Problems. Continuously Learn and Stay Updated															
CO-PO & PSO Mapping (Articulation Matrix):																
COs	PROGRAMME OUTCOMES												PSOs			
	01	02	03	04	05	06	07	08	09	10	11	12	01	02		
01	3	2	-	-	-	-	-	-	-	-	-	-	-	-		
02	2	3	3	-	2	-	-	-	-	-	-	-	-	-		
03	2	3	2	-	2	-	-	-	2	-	-	-	-	-		
04	2	1	-	-	-	2	1	-	-	-	-	2	-	-		
05	2	2	3	-	-	-	-	-	-	-	-	-	-	-		
06	1	-	-	-	-	-	-	-	-	-	-	2	-	-		
Avg.	2	2.2	2.6		2	2	1		2			2				
Syllabus																
UNIT I												(12 Hours)				
Introduction to Data Science – Evolution of Data Science – Data Science Roles – Stages in a Data Science Project – Applications of Data Science in various fields – Data Security Issues, Data Collection Strategies, Data Categorization: NOIR Topology.																
UNIT II												10 Hours				
Descriptive Statistics – Mean, Median, Standard Deviation, Skewness and Kurtosis. Data Pre-Processing Overview – Data Cleaning – Data Integration and Transformation – Data Reduction – Data Discretization.																
UNIT III												08 Hours				
Model Selection and Evaluation using Visualization:- Simple and Multiple Regression , Polynomial Regression, Ridge Regression , Box Plots, Residual Plot, Distribution Plot, Pivot Table.																
UNIT IV												10 Hours				
Introduction hypothetical testing:-NULL(H ₀) hypothesis and Alternative hypothesis, Type-I and Type-II error, Test statistics:- Z-statistics, T-statistics, Chi-Square Test, Correlation Statistics – ANOVA, correlation Analysis- Heat Map.																
UNIT V												6 Hours				
Model evaluation Metrics:- Regression metrics and classification metrics– Cross Validation – Overfitting – Under Fitting and Testing Multiple Parameters by using Grid Search.																
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs																

Text Books:

1. Cathy O'Neil and Rachel Schutt , *"Doing Data Science"*, O'Reilly, 2015.
2. Cathy O'Neil and Rachel Schutt , *"Doing Data Science"*, O'Reilly, 2015.

Reference Book:

1. Jojo Moolayil, *"Smarter Decisions : The Intersection of IoT and Data Science"*, PACKT, 2016.
2. Raj, Pethuru, *"Handbook of Research on Cloud Infrastructures for Big Data Analytics"*, IGI Global

Subject Code	Name of the Subject										L	T	P	C	
	Applied Statistics										3	1	0	3	
Course Educational Objectives															
Pre-Requisite: A student should have basic idea on programming and file systems.															
CEO 1															
CEO 2															
CEO 3															
CEO 4															
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Identify the role that statistics can play in the engineering problem-solving process, discuss the different methods that engineers use to collect data and, construct and interpret visual data displays.														
CO2	Compute and interpret the descriptive statistics, correlation coefficient and rank correlation coefficient, use simple linear regression model to engineering data.														
CO3	Explain various sampling methods, compute and explain point estimators and interval estimators for mean, variance and proportion														
CO4	Structure engineering decision-making problems as hypothesis tests, use z-test, t-test, chi-square and Ftest to test the statistical hypotheses, p-value approach for making decisions in hypothesis tests, explain and use the relationship between confidence interval and hypothesis tests.														
CO-PO & PSO Mapping:															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	2	1												
CO2	2	2	1												
CO3	2	2	2												
CO4	2	2	2												
Avg.	1.75	2	1.5												
SYLLABUS															
UNIT:1 (10 Hours)															
The Role of Statistics in Engineering : The Engineering Method and Statistical Thinking - Collecting Engineering Data - Basic Principles - Retrospective Study - Observational Study - Designed Experiments -Observing Processes Over Time - Mechanistic and Empirical Models															
Data Description and Representation: Collection of data- Classification and Tabulation of data - Stem-and-Leaf Diagrams - Frequency Distributions and Histograms - Box Plots - Time Sequence Plots - Probability Plots															
UNIT:2 (10 Hours)															
Descriptive Statistics: Measures of central Tendency-Measures of Dispersion- Skewness and Kurtosis.															
Correlation and Regression: Scatter Diagram – Types of Correlation – Karl Pearsons Coefficient of Correlation and Spearmen’s Rank Correlations- Method of Least Squares – Linear Regression.															

UNIT:3	(12 Hours)
Sampling: Different types of sampling - Sampling Distributions – Sampling Distribution of Mean. Point Estimation of Parameters: General Concepts of Point Estimation - Unbiased Estimators -Variance of a Point Estimator - Standard Error- Methods of Point Estimation (Method of Moments - Method of Maximum Likelihood). Statistical Intervals for a Single Sample: Confidence Interval on the Mean of a Normal Distribution with Variance Known - Confidence Interval on the Mean of a Normal Distribution with Variance Unknown - Confidence Interval on the Variance and Standard Deviation of a Normal Distribution - A Large-Sample Confidence Interval for a Population Proportion.	
UNIT:4	(12 Hours)
Tests of Hypotheses for a Single Sample: Tests of Statistical Hypotheses - General Procedure for Hypothesis Testing –Tests on the Mean of a Normal Distribution with Variance Known - Tests on the Mean of a Normal Distribution with Variance Unknown - Tests on the Variance and Standard Deviation of a Normal Distribution . Statistical Inference for Two Samples: Inference For a Difference in Means of Two Normal Distributions with Variances Known - Inference For a Difference in Means of Two Normal Distributions with Variances Unknown -Inference on the Variances of Two Normal Distributions – Inference on Two Population Proportions.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Douglas C. Montgomery and George C. Runger. <i>Applied Statistics and Probability for Engineers</i> , (3rdEdn), John Wiley and Sons, Inc., New York, 2003. 2. Robert H. Carver and Jane Gradwohl Nash. <i>Doing Data Analysis with SPSS Version 18.0</i> , (Indian Edition), Cengage Learning, New Delhi, 2012. 3. Richard A. Johnson and C.B.Gupta, <i>Probability and Statistics for Engineers</i> , (7thEdn.), Pearson Education, Indian Impression 2006.	
Reference Books: 1. Mohammed A.Shayib. <i>Applied Statistics, First Edition. EBook, Bookboon.com</i> 2013. 2. Peter R.Nelson, Marie Coffin, Copeland Kanen, A.F. <i>Introductory Statistics for Engineering Experimentation, Elsevier Science and Technology Books, New York, 2003.</i> 3. Sheldon M. Ross, <i>Introduction to Probability and Statistics, (3rdEdn), Elsevier Science and Technology Books, New York, 2004.</i> 4. T.T.Soong, <i>Fundamentals of Probability and Statistics for Engineers, John Wiley and Sons, Ltd., New York, 2004.</i> 5.J.P.Marques de Sá , <i>Applied Statistics using SPSS, STATISTICA, MATLAB and R, (2ndEdn.), Springer Verlag, Heidelberg, 2007.</i>	
Online resources ➤ http://apus.libguides.com ➤ www.mii.lt ➤ www.sosmath.com ➤ www.macalester.edu ➤ www.nptel.iitm.ac.in ➤ www.purplemath.com	

Pre-requisites (if any):		Knowledge of Basic Mathematics & Basic Electrical and Electronics Engineering.
Course Objectives:		
–	To understand digital logic fundamentals, including binary systems and Boolean algebra.	
–	To design and analyze combinational and sequential circuits, including logic gates, flip-flops, and counters.	
–	To apply digital electronics in real-world applications, using simulation tools and programmable logic devices (PLDs).	
Course Content		
Unit 1		(10 Hours)
Number Systems and Codes: Binary, Octal, Hexadecimal, and Decimal Number System and their Conversion; Representation of Signed Binary and Floating-Point Number; Binary Arithmetic using 1’s and 2’s Complements, Binary Codes - BCD Code, Gray Code, ASCII Character Code.		
Boolean Algebra and Logic Gates: Axioms and Laws of Boolean Algebra; Reducing Boolean Expressions; Logic Levels and Pulse Waveforms; Logic Gates; Boolean Expressions and Logic Diagrams; Canonical and Standard Forms.		
Unit 2		(12 Hours)
Gate-level Minimization: K-maps - Two, Three, and Four Variable K-maps, Don’t-Care Conditions; NAND and NOR Implementation; Other Two-Level Implementations, Exclusive-OR Function.		
Combinational Logic: Combinational Circuits; Analysis Procedure; Design Procedure; Adders; Subtractors; Parallel Binary Adders; Binary Adder-Subtractor; Binary Multiplier; Magnitude Comparator; Decoders; Encoders; Multiplexers; De-multiplexers.		
Unit 3		(8 Hours)
Synchronous Sequential Logic: Sequential Circuits; Storage Elements: Latches, Flip-Flops, Master-Slave Flip-Flop; Conversion of Flip-Flops; Analysis of Clocked Sequential Circuits; Mealy and Moore Models of Finite State Machines; Design Procedure.		
Unit 4		(8 Hours)
Registers and Counters: Shift Registers; Data Transmission in Shift Registers; SISO, SIPO, PISO, and PIPO Shift Registers; Counters; Asynchronous Counters; Design of Asynchronous Counters; Synchronous Counters; Design of Synchronous Counters; Ring Counter.		
Unit	(6 Hours)	
Memory and Programmable Logic: Introduction; Random-Access Memory; Memory Decoding; Error Detection and Correction; Read-Only Memory; Programmable Logic Array; Programmable Array Logic; Sequential Programmable Devices.		
Text Books:		
[1]	M. Morris Mano, and Michael D. Ciletti, <i>Digital Design: With an Introduction to the Verilog HDL, VHDL, and SystemVerilog</i> . Pearson Education, Sixth Edition, 2017.	
[2]	A. Anand Kumar, <i>Fundamentals of Digital Circuits</i> . PHI Learning Pvt. Ltd., New	

	Delhi, Fourth Edition, 2016.
[3]	R. P. Jain, <i>Modern Digital Electronics</i> . Tata McGraw-Hill Education Pvt. Ltd., Fourth Edition, 2009.

[3] R. P. Jain, *Modern Digital Electronics*. Tata McGraw-Hill Education Pvt. Ltd., Fourth Edition, 2009.

Reference Books:

- [1] John P. Uyemura, *A First Course in Digital Systems Design: An Integrated Approach*. Thomson Press (India) Ltd., India Edition, 2002.
- [2] William H. Gothmann, *Digital Electronics: An Introduction to Theory and Practice*, PHI Learning Pvt. Ltd., New Delhi, Second Edition, 2006.
- [3] D.V. Hall, *Digital Circuits and Systems*. Tata McGraw-Hill Education Pvt. Ltd., 1989.
- [4] Charles H. Roth, *Digital System Design using VHDL*. Tata McGraw-Hill Education Pvt. Ltd., Second Edition, 2012.

- [1] John P. Uyemura, *A First Course in Digital Systems Design: An Integrated Approach*. Thomson Press (India) Ltd., India Edition, 2002.
- [2] William H. Gothmann, *Digital Electronics: An Introduction to Theory and Practice*, PHI Learning Pvt. Ltd., New Delhi, Second Edition, 2006.
- [3] D.V. Hall, *Digital Circuits and Systems*. Tata McGraw-Hill Education Pvt. Ltd., 1989.
- [4] Charles H. Roth, *Digital System Design using VHDL*. Tata McGraw-Hill Education Pvt. Ltd., Second Edition, 2012.

Course Outcomes:

At the end of this course, students will be able to

- O1: Define and analyze the number systems and conversions.
- O2: Perform binary arithmetic with 1's and 2's complements and understand binary codes.
- O3: Apply Boolean algebra laws to simplify expressions and design logic circuits.
- O4: Use K-maps for logic minimization and design combinational circuits.
- O5: Analyze and design synchronous sequential circuits.
- O6: Design shift registers, and counters, and understand RAM, ROM, and PLDs.

CO-PO & PSO matrices of Course

[illegible]

DIGITAL ELECTRONICS LABORATORY		0-0-2
Pre-requisites (if any):	Knowledge of Basic Electrical and Electronics Engineering.	
Course Objectives:		
–	To gain hands-on experience with basic logic gates and combinational circuits.	
–	To implement and test combinational and sequential circuits.	
–	To utilize simulation tools for circuit design and testing.	
Course Content		
List of Experiments:		
(At least 10 experiments should be done, Experiments No. 1 and 2 are compulsory, and out of the balance 8 experiments at least 3 experiments have to be implemented through both Verilog/VHDL and hardware implementation as per the choice of the student totaling to 6 and the rest 2 can be either through Verilog/VHDL or hardware implementation.)		
1. Digital Logic Gates: Investigate the logic behavior of AND, OR, NAND, NOR, EX-OR, EXNOR, Inverter, and Buffer Gates, use of Universal NAND/NOR Gate. 2. Gate-Level Minimization: Two-level and multi-level implementation of Boolean functions. 3. Combinational Circuits: Design, assemble, and test: adders and subtractors, code converters, gray code to binary, and 7-segment displays. 4. Design, implement, and test a given design example with (i) NAND Gates only, (ii) NOR Gates only, and (iii) using the minimum number of Gates. 5. Design with Multiplexers and De-multiplexers. 6. Flip-Flop: Assemble, test, and investigate the operation of SR, D & JK flip-flops. 7. Shift Registers: Design and investigate the operation of all types of shift registers with parallel load. 8. Counters: Design, assemble, and test various ripple and synchronous counters - decimal counter, Binary counter with the parallel load. 9. Memory Unit: Investigate the behavior of the RAM unit and its storage capacity - 16×4 RAM: testing, simulating, and memory expansion. 10. Clock-Pulse Generator: Design, implement, and test. 11. Parallel Adder and Accumulator: Design, implement, and test. 12. Binary Multiplier: Design and implement a circuit that multiplies 4-bit unsigned numbers to produce an 8-bit product. 13. Verilog/VHDL simulation and implementation of Experiments listed in Sl. No. 3 to 12.		

Course Outcomes:

At the end of this course, students will be able to

CO1: Investigate the behavior and application of various digital logic gates.

CO2: Implement two-level and multi-level Boolean function minimization using universal gates like NAND and NOR.

CO3: Design, assemble, and test combinational circuits.

CO4: Assemble, test, and investigate flip-flops and shift registers, and design various types of counters

CO5: Investigate RAM behavior and design memory units.

CO6: Use Verilog/VHDL for simulation and implementation of combinational and sequential circuit designs.

CO-PO & PSO matrices of Course

[illegible]

Subject Code		Name of the Subject		L	T	P	C	
		Intermediate Communication Skills and Critical Thinking		3	0	0	3	
Course Educational Objectives								
CE01	To develop the communication skills and soft skills of the students.							
CE02	To enhance employability and entrepreneurial skills.							
CE03	To promote critical thinking.							
Course Outcomes: Upon successful completion of this course, students should be able to:								
CO1	Recognise the importance of LSRW skills.							
CO2	Develop effective listening skills.							
CO3	Demonstrate public speaking skills.							
CO4	Distinguish professional writing from general writing.							
CO5	Develop effective reading skills.							
CO6	Execute critical thinking.							

CO-PO & PSO Mapping

COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1										3					
CO2											2				
CO3										3					
CO4									3						
CO5										3					
CO6										3					
Avg.									3	3	2				

SYLLABUS

UNIT-1 Comprehensive Listening Skills	(4 Hours)
1.1 What and why 1.2 Techniques for improving listening skills	
UNIT-2 Public Speaking Skills	(4 Hours)
2.1 Public speaking; what and why 2.2 Overcoming stage fear 2.3 Techniques for effective speaking skill	
UNIT- 3 Stages of Writing	(4 Hours)
3.1 Pre-Writing- selecting a topic 3.2 Writing- Free Writing, Drafting 3.3 Re-writing- Revision, editing, proof Reading	
UNIT- 4 Effective Reading Skills	(3 Hours)
4.1 What is Reading Comprehension 4.2 Tips for effective reading 4.3 Techniques and types.	
UNIT-5 Critical Thinking Skills	(4 Hours)
5.1 What and Why 5.2 Critical thinking vs. Lateral thinking 5.3 Ways to improve Critical Thinking	

Teaching Methods: Chalk & Board/PPT/Video Lectures

Reference Books:

1. Basic Communication Skills for Technology by Andrea J. Rutherford, Pearson Education Asia, Patparganj, New Delhi, 2001.
2. How to Read better and Faster by Norman Lewis, 4th Edition, Publisher: Crowell, 1978.
3. The Kabuliwala and other stories by Rabindranath Tagore, Maple Classics, 2019.
4. The Gift of Magi by O Henry, Candlewick Press, 2008.

Subject Code	Name of the Subject (3 rd Sem)	L	T	P	C	
	Intermediate Communication skills and Critical thinking (Lab)	0	0	2	1	

Course Educational Objectives

CEO1	To develop communication skills and soft skills of the students.
CEO2	To enhance employability and entrepreneurial skills.
CEO3	To promote critical thinking.

Course Outcomes: Towards the end of the course, the students will be able to:

CO1	Recognise the importance of LSRW skills
CO2	Develop effective listening skills
CO3	Demonstrate public speaking skills
CO4	Distinguish professional writing from general writing
CO5	Develop effective reading skills
CO6	Execute critical thinking

CO-PO & PSO Mapping

Cos	PROGRAMME OUTCOMES														
	1	2	3	4	5	6	7	8	9	10	11	12			
CO1										3					
CO2											3				
CO3											3				
CO4											3				
CO5											3				
CO6									3			2			
AVG									3		3	2			

SYLLAYBUS

Comprehensive Listening (2 hours)

Activity- Summarizing a Passage

Comprehensive Listening (2 hours)

Activity- Listening to a short video/ Ted Talk and giving a brief.

Oral Presentation (2 hours)

Activity- Presenting seminar paper in a team and presenting to the class

Individual Oral Presentation (2 hours)

Activity- Preparing a seminar paper individually and presenting.

Persuasive Speaking (2 hours)

Activity- Individual/ Group Debate.

Situational Interviews (2 hours)

Activity- Answering hypothetical question in a mock interview	
Video Resume	(2 hours)
Activity- Preparing a digital resume	
Professional e-mail	(2 hours)
Activity- Writing e-mail for professional purpose	
Minutes of the Meeting	(2 hours)
Activity- Preparing minutes of mock meetings.	
Memo Writing	(2 hours)
Writing Memos for internal organization	
Vocabulary Exercise	(2 hours)
Activity- Playing word games in small groups.	
Intermediate Reading Comprehension	(2 hours)
Activity-Reading an Intermediate level passage/ poems and answer questions.	
Preparing a Sales Pitch	(2 hours)
Activity- Prepare a presentation on any product (real/imaginary) of choice and present a sales pitch	
Critical Thinking Activity	(2 hours)
Activit- critically analysis of Kabuliwala by Rabindranath Tagore OR The Gift of Magi by O Henry	
Critical Thinking	(2 hours)
Activity 2- Critically Analyse Stopping by the Woods in a Snowy Evening by Robert Frost OR Ozymandias by P.B Shelly.	
Reference Books	
1. An Introduction to Professional English and Soft Skills by B. K. Das et al., Cambridge University Press, 2009. 2. Communicative English for Engineers and Professionals by Nitin Bhatnagar and Mamta Bhatnagar. Published by DK/Pearson, 2010. 3. Practical English Usage . Michael Swan, OUP, 1995	

IV Semester [Second Year]
B. Tech in Computer Science and Engineering (Data Science) (2023 Admission)

S N	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	PC	23BCDPC24001	Python Programming for Data Science	3	0	0	3	40	60
2	PC	23BCDPC24002	Computer Organization & Architecture	3	0	0	3	40	60
3	PC	23BCDPC24003	Design and Analysis of Algorithms	3	0	0	3	40	60
4	PC	23BCDPC24004	Operating Systems	3	0	0	3	40	60
5	PE	23BCDPE24011 / 23BCDPE24012	PE-II (From Bucket)	3	0	0	3	40	60
6	HS	23BBSHS24001	English Learning -II (Advance communication skills and professional Ethics)	1	0	0	1		
			TOTAL	16	0	0	16	200	300
PRACTICAL/SESSIONAL									
7	PC	23BCDPC24101	Python Programming Lab	0	0	2	1	30	20
8	PC	23BCDPC24103	Design and Analysis of Algorithms Lab	0	0	2	1	30	20
9	PC	23BCDPC24105	Data Analytics Tool Laboratory	0	0	2	1	30	20
10	HS	23BBSHS24101	English Learning -II (Advance communication skills and professional Ethics lab)	0	0	0	1	30	20
11	PST	23BCSLP24101	Learning Project-II	0	0	0	1	50	0
			TOTAL	0	0	6	5	170	80
			TOTAL	16	0	6	21	370	380

Professional Elective Bucket

Sl .	Sem	Professional Elective II
1	23BCDP E24011	Predictive Modeling and Analytics
2	23BCDP E24012	Statistical Inference for Data Science

Subject Code		Name of the Subject						L	T	P	C	QP			
		Python Programming for Data Science						3	1	0	4	A			
Course Educational Objectives															
CEO1	To understand the basics of programming using Python.														
CEO2	To Construct and execute basic programs in Python.														
CEO3	To implement database connectivity, CGI, Multithreading and web development in python.														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	Understand the basic concept of Python like data types and syntax of programming.														
CO2	Use Python programming with user-defined & built-in methods, and objects of Python.														
CO3	Applications of Object Oriented Concepts and exception handling.														
CO4	To apply data manipulation, analysis, and visualization using NumPy, Pandas, and Matplotlib packages in Python.														
CO5	Apply Python libraries such as NumPy, Pandas, and Matplotlib to manipulate, analyze, and visualize datasets, demonstrating their ability to handle real-world data science tasks.														
CO 6	Evaluate various machine learning models implemented in Python by using appropriate metrics and techniques, and justify the selection of the most suitable model for a given data science problem														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES														
	1	2	3	4	5	6	7	8	9	10	11	12			
CO1	3	1			1							1			
CO2	2	3	1	3	2							1			
CO3	2	3	3	3	2				1		1	2			
CO4	2	3	3	3	2				1		1	2			
CO5	2	3	3	3	3										
CO6	2	3	3	3	3										
Avg.	2.1	2.6	2.1	3	2.1				1		1	1.5			
SYLLABUS															
UNIT:1 (08 Hours)															
Introduction: Installation, First Python Program: Interactive Mode Programming, Script Mode Programming; Identifiers, Reserved Words, Lines and Indentation, Multi-Line Statements, Quotation & Comments; Assigning Values to Variables, Multiple Assignment.															
Standard Data Types: Numbers, Strings, Lists, Tuples, Dictionary; Data Type Conversion; Basic Operators: Arithmetic, Comparison, Assignment, Bitwise; Operators: Logical, Membership, Identity; Operators Precedence; Python Numbers & Mathematical functions, Python Strings.															
UNIT:2 (10 Hours)															
Python Program Flow Control : Conditional blocks using if, else and elif, Simple for loops in python, For loop using ranges, string, list and dictionaries, Use of while loops in python, Loop manipulation using pass, continue, break and else, Programming using Python conditional and loops block															
Python Functions, Modules And Packages: Organizing python codes using functions,															

Organizing python projects into modules, Importing own module as well as external modules, Understanding Packages, Powerful Lambda function in python, Programming using functions, modules and external packages	
Python String, List And Dictionary Manipulations : Building blocks of python programs, Understanding string in build methods, List manipulation using in build methods, Dictionary manipulation, Programming using string, list and dictionary in build functions	
UNIT:3	(12 Hours)
Python Object Oriented Programming – OOPS: Concept of class, object and instances, Constructor, class attributes and destructors, Real time use of class in live projects, Inheritance , overlapping and overloading operators, Adding and retrieving dynamic attributes of classes , Programming using OOPS support.	
Python Exception Handling : Avoiding code breaks using exception handling, Handling ZeroDivisionError, TypeError, and ValueError.	
Math Module: importing Math module, random(), randint(), uniform(), randrange(), and choice().	
Python Libraries for Data Science: Introduction to data science libraries in Python: NumPy, Pandas, and Matplotlib.	
Numpy: ndarray object, indexing, and slicing. Data types and data structures in NumPy, properties and functions: ones(), zeros(), empty(), shape, reshape(), copy(), view(), concatenate (), sort(), Numpy array, operations: max(), min(), sum(), prod(), broadcasting.	
UNIT:4	(15 Hours)
Data Manipulation: Pandas: Introduction, installation. Series, labels, data frame, Load files into Data Frame: reading csv and json files. Finding the relationship: corr().	
UNIT:5	(15 Hours)
Data Visualization: Introduction to Introduction to Matplotlib, Installation and Setup, Basic Plotting: Plotting simple graphs using plt.plot(), Line styles, colors, and markers, Adding labels, title, and grid. Different Types of Plots: Scatter plots, Bar plots, Histograms, Box plots, Pie charts, Contour plots, 3D plots	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books:	
1. “Python: The Complete Reference”, Martin C. Brown, McGraw-Hill/Osborne Media. 2. "Python Data Science Handbook: Essential Tools for Working with Data" by Jake VanderPlas, O'Reilly Media, 2016. 3. "NumPy Beginner's Guide" by Ivan Idris, Packt Publishing, 2015. 4. "Matplotlib for Python Developers" by Sandro Tosi, Packt Publishing, 2018.	
Reference Books:	
1. “Expert Python Programming”, Tarek Ziad, Packt Publishing. 2. “Python Essential Reference, David Beazley”, Developer's Library, Sams Publishing.	

Subject Code		Name of the Subject								L	T	P	C			
		Computer Organization and Architecture								3	0	0	3			
Course Educational Objectives																
CEO1	Identify the functional units in a digital computer system,															
CEO2	Distinguish between the various ISA styles and trace the execution sequence of an instruction through the processor.															
CEO3	Compare different approaches used for implementing a functional unit and evaluate different computer systems based on performance metrics.															
CEO4	Provide an outline of working principles of components of computer.															
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>																
CO1	Explain and illustrate the execution cycle of a program.															
CO2	Classify the components of digital computer and designing of control and memory units with performance evaluation.															
CO3	Compute different types of binary arithmetic operation. .															
CO4	Identify the characteristics of memories and explain memory and IO devices working process.															
CO5	Classify the concept of Parallelism															
CO6	Explain memory and types and interrupts															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	1	1			1											
CO2	2	3			3											
CO3	3	2			2											
CO4	1	1			1											
CO5	2	2			1											
CO6	1	3			2											
Avg.	1.6	2			1.6											
SYLLABUS																
UNIT:1 (12 Hours)																
Functional Units of a computer, Operational concepts, Bus structures, Memory locations and addresses, Memory operations, Instructions, Instruction sequencing, addressing modes, Problem solving, Introduction to Microprocessor, Introduction to Assembly language, writing of assembly language programming, ARM Processor: The thumb instruction set, Processor and CPU cores, Instruction Encoding format, Memory load and Store instruction in ARM, Basics of IO operations.																
UNIT: 2 (12 Hours)																
Addition and subtraction of Signed numbers, Problem solving, Design of fast adders, Ripple carry adder and Carry look ahead adder, Multiplication of positive numbers, Problem Solving, Signed operand multiplication, Problem solving, Fast multiplication- Bit pair recoding of Multipliers, Problem Solving, Carry Save Addition of summands - Problem Solving, Integer division – Restoring Division - Solving Problems, Non-Restoring Division- Solving Problems, Floating point numbers and operations- Solving Problems.																

UNIT:3	(12 Hours)
Fundamental concepts of basic processing unit, Performing ALU operation, Execution of complete instruction, Branch instruction, Multiple bus organization, Hardwired control, Generation of control signals, Micro-programmed control, Microinstruction, Micro-program Sequencing, Micro instruction with Next address field, Basic concepts of pipelining, Pipeline Performance, Pipeline Hazards-Data hazards, Methods to overcome Data hazards, Instruction Hazards, Hazards on conditional and Unconditional Branching, Control hazards, Influence of hazards on instruction sets.	
UNIT:4	(12 Hours)
Parallelism, Need, types of Parallelism, applications of Parallelism, Parallelism in Software, Instruction level parallelism, Data level parallelism, Challenges in parallel processing, Architectures of Parallel Systems - Flynn's classification, SISD, SIMD, MIMD, MISD, Hardware multithreading, Coarse Grain parallelism, Fine Grain parallelism, Uni-processor and Multiprocessors, Multi-core processors, Memory in Multiprocessor Systems, Cache Coherency in Multiprocessor Systems, MESI protocol for Multiprocessor Systems.	
UNIT:5	(10 Hrs)
Memory systems -Basic Concepts, Memory hierarchy, Memory technologies, RAM, Semiconductor RAM, ROM, Types, Speed, size cost, Cache memory, Mapping Functions, Replacement Algorithms - Problem Solving, Virtual Memory, Performance considerations of various memories, Input Output Organization, Need for Input output devices, Memory mapped IO, Program controlled IO, Interrupts-Hardware, Enabling and Disabling Interrupts, Handling multiple Devices.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky and Naraig Manjikian, "Computer Organization and Embedded Systems", Sixth Edition, Tata McGraw Hill, 2012. 2. David A. Patterson and John L. Hennessy, "Computer Organization and Design: The Hardware/Software Interface", Fourth Edition, Morgan Kaufmann / Elsevier, 2009. 3. Kai Hwang and F.A. Briggs, "Computer Architecture and Parallel Processing", McGraw Hill.	
Reference Books: 4. M. Morris Mano, "Computer System Architecture", PHI 5. William Stallings, "Computer Organization and Architecture – Designing for Performance", Sixth Edition, Pearson Education, 2003. 6. John P. Hayes, "Computer Architecture and Organization", Third Edition, Tata McGraw Hill, 1998. 7. John L. Hennessy and David A. Patterson, "Computer Architecture – A Quantitative Approach", Morgan Kaufmann / Elsevier Publishers, Fifth Edition, 2012.	

Subject Code	Name of the Subject	L	T	P	C										
	Design and Analysis of Algorithms	3	0	0	3										
Course Educational Objectives															
CEO1	Analyze the asymptotic performance of algorithms														
CEO2	Demonstrate familiarity with major algorithms														
CEO3	Apply important algorithmic design paradigms and methods of analysis														
CEO4	Synthesize efficient algorithms in common engineering design situations														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	Apply efficient algorithms to reduce space and time complexity of both recurrent and														
CO2	Solve problems using divide and conquer approaches														
CO3	Apply greedy and dynamic programming types techniques to solve polynomial time														
CO4	Create exponential problems using backtracking and branch and bound approaches														
CO5	Interpret various approximation algorithms and interpret solutions to evaluate P type,														
CO6	Create algorithms that are efficient in space and time complexities by using divide														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	1	1												
CO2	2	2	2												
CO3	3	1	1												
CO4	3	1	1												
CO5	2	2	3												
CO6	1	3	2												
Avg.	2	1.6	1.6												
SYLLABUS															
Unit I (15Hrs)															
Introduction-Algorithm Design, Fundamentals of Algorithms, Correctness of algorithm, Time complexity analysis, Insertion sort-Line count, Operation count, Algorithm Design paradigms, Designing an algorithm, And its analysis-Best, Worst and Average case, Asymptotic notations Based on growth functions, O,O,Θ, ω, Ω, Mathematical analysis, Induction, Recurrence relations- Solution of recurrence relations, Substitution method- Solution of recurrence relations, Recursion tree- Solution of recurrence relations.															
Unit -II (15Hrs)															
Introduction-Divide and Conquer, Maximum Subarray Problem, Binary Search, Complexity of binary search, Merge sort, Time complexity analysis, Quick sort and its Time complexity analysis, Best case, Worst case, Average case analysis, Strassen's Matrix multiplication and its recurrence relation, Time complexity analysis of Merge sort, Largest sub-array sum, Time complexity analysis of Largest subarray sum, Master Theorem Proof, Master theorem examples, Finding Maximum and Minimum in an array, Time complexity analysis-Examples, Algorithm for finding closest pair problem, Convex Hull problem.															
Unit – III (10Hrs)															
Introduction-Greedy and Dynamic Programming, Examples of problems that can be solved by using greedy and dynamic approach, Huffman coding using greedy approach, Comparison of brute force and Huffman method of encoding, Knapsack problem using greedy approach, Complexity derivation of knapsack using greedy, Tree traversals, Minimum spanning tree –															

greedy Kruskal's algorithm – greedy, Minimum spanning tree - Prims algorithm, Introduction to dynamic programming, 0/1 knapsack problem, Complexity calculation of knapsack problem, Matrix chain multiplication using dynamic programming, Complexity of matrix chain multiplication, Longest common subsequence using dynamic programming, Explanation of LCS with an example, Optimal binary search tree (OBST) using dynamic programming, Explanation of OBST with an example.	
Unit – IV	(10 Hrs)
Introduction to backtracking - branch and bound, N queen's problem – backtracking, Sum of subsets using backtracking, Complexity calculation of sum of subsets, Graph introduction, Hamiltonian circuit – backtracking, Branch and bound - Knapsack problem, Example and complexity calculation, Differentiate with dynamic and greedy, Travelling salesman problem using branch and bound, Travelling salesman problem using branch and bound example, Time complexity calculation with an example, Graph algorithms, Depth first search and Breadth first search, Shortest path introduction, Floyd-Warshall Introduction, Floyd-Warshall with sample graph, Floyd-Warshall complexity.	
Unit – V	(10 Hrs)
Introduction to randomization and approximation algorithm, Randomized hiring problem, Randomized quick sort, Complexity analysis, String matching algorithm – Examples, Rabin Karp algorithm for string matching - Example discussion, Approximation algorithm, Vertex covering, Introduction Complexity classes, P type problems, Introduction to NP type problems, Hamiltonian cycle problem, NP complete problem introduction, Satisfiability problem, NP hard problems – Examples.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Book:	
1. <i>Introduction to Algorithms</i> , T.H. Cormen, C.E. Leiserson, R.L. Rivest, C. Stein, 2nd Edition, PHI Learning Pvt. Ltd.	
2. <i>Fundamentals of Algorithm</i> , Horowitz & Sahani, 2nd Edition, Universities Press.	
Reference Book:	
1. <i>Algorithms, Design and Analysis</i> , H. Bhasin, First Edition, Oxford University	
2. <i>Design and Analysis of Algorithm</i> , S. Sridhar, Oxford University Press	
3. <i>Algorithms</i> , Sanjay Dasgupta, Umesh Vazirani, McGraw-Hill Education.	

Subject Code	Name of the Subject	L	T	P	C										
	Operating System	3	0	2	4										
Course Educational Objectives															
CEO1	Provide an insight into the basic organization of computer systems.														
CEO2	Familiarize students with various components of an Operating System														
CEO3	Focus on fundamental problems and optimal solutions for resource management														
CEO4	Understand different possible solutions for handling deadlock situation.														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	Identify the need of an Operating system														
CO2	Know the Process management functions of an Operating system														
CO3	Understand the need of Memory Management functions of an Operating system														
CO4	Find the significance of Device management role of an Operating system														
CO5	Recognize the essentials of File Management part of an Operating system														
CO6	Gain an insight of Importance of an Operating system through practical														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	2	2	1				1								
CO2	2	1	2				2								
CO3	1	1	3				3								
CO4	2	3	2				1								
CO5	3	1	1				1								
CO6	1	2	1				1								
Avg.	1.8	1.6	1.6				1.5								
SYLLABUS															
UNIT:1 (14 Hours)															
Operating System Objectives and functions, Gaining the role of Operating systems, The evolution of operating system, Major achievements, Understanding the evolution of Operating systems from early batch processing systems to modern complex systems, OS Design considerations for Multiprocessor and Multicore, Understanding the key design issues of Multiprocessor Operating systems and Multicore Operating systems, PROCESS CONCEPT– Processes, PCB, Understanding the Process concept and Maintenance of PCB by OS, Threads – Overview and its Benefits, Understanding the importance of threads, Process Scheduling : Scheduling Queues, Schedulers, Context switch, Understanding basics of Process scheduling, Operations on Process – Process creation, Process termination, Understanding the system calls – fork(),wait(),exit(),Inter Process communication : Shared Memory, Message Passing, Pipe(),Understanding the need for IPC, PROCESS SYNCHRONIZATION: Background, Critical section Problem, Understanding the race conditions and the need for the Process synchronization															

UNIT:2	(12 Hours)
PROCESS SYNCHRONIZATION: Peterson's solution, Synchronization Hardware, Understanding the two-process solution and the benefits of the synchronization hardware, Process synchronization: Semaphores, usage, implementation, Gaining the knowledge of the usage of the semaphores for the Mutual exclusion mechanisms, Classical Problems of synchronization – Readers writers problem, Bounded Buffer problem, Good understanding of synchronization mechanisms, Classical Problems of synchronization – Dining Philosophers problem (Monitor) Understanding the synchronization of limited resources among multiple processes, CPU SCHEDULING: FCFS,SJF,Priority, Understanding the scheduling techniques, CPU Scheduling: Round robin, Multilevel queue Scheduling, Multilevel feedback Scheduling, Understanding the scheduling techniques, Real Time scheduling: Rate Monotonic Scheduling and Deadline Scheduling, Understanding the real time scheduling, DEADLOCKS: Necessary conditions, Resource allocation graph, Deadlock prevention methods, Understanding the deadlock scenario, Deadlocks :Deadlock Avoidance, Detection and Recovery, Understanding the deadlock avoidance, detection and recovery mechanisms	
UNIT:3	(14 Hours)
MEMORY MANAGEMENT: Memory Management: Logical Vs Physical address space, Swapping, Understanding the basics of Memory management, Contiguous Memory allocation – Fixed and Dynamic partition, getting to know about Partition memory management and issues: Internal fragmentation and external fragmentation problems, Strategies for selecting free holes in Dynamic partition, Understanding the allocation strategies with examples, Paged memory management, Understanding the Paging technique. PMT hardware mechanism, Structure of Page Map Table, Understanding the components of PMT, Example : Intel 32 bit and 64 –bit Architectures, Understanding the Paging in the Intel architectures, Example : ARM Architectures, Understanding the Paging with respect to ARM, Segmented memory management, Understanding the users view of memory with respect to the primary memory, Paged segmentation Technique, Understanding the combined scheme for efficient management	
UNIT:4	(14 Hours)
VIRTUAL MEMORY– Background, Understanding the need of demand paging, VIRTUAL MEMORY – Basic concepts – page fault handling, Understanding, how an OS handles the page faults, Performance of Demand paging, Understanding the relationship of effective access time and the page fault rate, Copy-on write, Understanding the need for Copy-on write, Page replacement Mechanisms: FIFO, Optimal, LRU and LRU approximation Techniques, Understanding the Pros and cons of the page replacement techniques, Counting based page replacement and Page Buffering Algorithms, To know on additional Techniques available for page replacement strategies, Allocation of Frames - Global Vs Local Allocation, Understanding the root cause of the Thrashing, Thrashing, Causes of Thrashing, Understanding the Thrashing, Working set Model, Understanding the working set model for controlling the Working set Model.	
UNIT:5	(14 Hours)
STORAGE MANAGEMENT: Mass storage structure – Overview of Mass storage structure – Magnetic Disks, Understanding the Basics in storage management, Disk Scheduling, Understanding the various scheduling with respect to the disk, FILE SYSTEM INTERFACE: File concept, File access methods, Understanding the file basics, File sharing and Protection,	

Emphasis the need for the file sharing and its protection, FILE SYSTEM IMPLEMENTATION: File system structure, To get the basic file system structure, Directory Implementation, Understanding the various levels of directory structure, FILE SYSTEM IMPLEMENTATION :Allocation methods, Understanding the pros and Cons of various disk allocation methods, FILE SYSTEM IMPLEMENTATION :Free space Management, Understanding the methods available for maintaining the free spaces in the disk, Swap space Management, Understanding the Low-level task of the OS.

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Books/References:

1. *Operating System Principles*, Abraham Silberchatz, Peter B. Galvin, Greg Gagne 8th Edition, Wiley Student Edition.
2. *Operating systems - Internals and Design Principles*, W. Stallings, 6th Edition, Pearson.M. G. Say, "Performance and design of AC machines", CBS Publishers, 2002.
3. *Modern Operating Systems*, Andrew S Tanenbaum 3rd Edition PHI.
4. *Operating Systems A concept - based Approach*, 2nd Edition, D. M. Dhamdhare, and TMH.
5. *Principles of Operating Systems*, B. L. Stuart, Cengage learning, India Edition.
6. *Operating Systems*, A. S. Godbole, 2nd Edition, TMH
7. *An Introduction to Operating Systems*, P.C.P. Bhatt, PHI.
8. *Operating Systems*, S, Haldar and A. A. Arvind, Pearson Education.
9. *Operating Systems*, R. Elmasri, A. G. Carrick and D. Levine, McGraw Hill.
10. *Operating Systems in depth*, T. W. Doeppner, Wiley.

Subject Code		Name of the Subject						L	T	P	C				
		Predictive Modelling and Analytics						3	0	1	4				
Course Educational Objectives															
CEO1: Students will learn the basic concepts data preparation															
CEO2: Recognize the limitations and challenges associated with data classification															
CEO3: The students will understand the basics of Data prediction.															
CEO4: They will also learn and will be able to apply different machine learning models to various datasets and Data visualization tools.															
Course Outcomes: : Upon successful completion of this course, students should be able to:															
CO1	Understand the basics of predictive analytics and summarize Data, Categorize Models and techniques														
CO2	Apply Decision tree, Support Vector Machine for Data Classification														
CO3	Apply methods such as Navie Bayes Markov Model, Linear regression , Neural Networks to boost prediction accuracy for data classification.														
CO4	Develop predictive models for various real-time applications and analyze , visualize predictive models results using data visualization.														
CO5	Analyze datasets and evaluate the performance of predictive models using Python, applying various machine learning techniques and metrics such as accuracy, precision, recall, and F1-score.														
CO6	Create and apply predictive models using Python, incorporating appropriate algorithms (e.g., linear regression, decision trees, random forests) to solve real-world analytics problems, and optimize models through hyperparameter tuning and cross-validation.														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3					2		1							
CO2	3	2	3	1											
CO3	3	2	3	1											
CO4	3	3	2		2										
CO5	3	3	2												
Avg.	3	2.5	2.5	1	2										
SYLLABUS															
UNIT:1 Data Preparation (12 Hours)															
Introduction – Predictive Analytics in the wild – Exploring Data Types and associated techniques – Complexities of data – Applying Models: Models and simulation, Categorizing Models, describing, summarizing data and decisions – Identify similarities in Data: Data clustering, converting raw data into Matrix, Identify K-groups in Data.															
UNIT:2 Data Classification (12 Hours)															
Background – Exploring data classification process- using Data Classification to predict the future: Decision Tree, Algorithm for generating Decision trees, Support Vector Machine,															

UNIT:3 Data Classification Ensemble Methods to Boost Prediction Accuracy: Navie Bayes Classification Algorithm, The Markov Model, Linear Regression, Neural Networks – Deep Learning.
UNIT:4 Data Prediction (12 Hours) Adopt Predictive analytics – Processing data: Identifying, Cleaning, generating, reducing dimensionality of data – structuring data – build predictive model: Develop and test the model.
UNIT:5 Data Visualization (12 Hours) Introduction to visualization tools – Evaluate the data – visualize Models Analytical Results: Hidden grouping, data classification results, outliers, decision trees, prediction – Novel Visualization in Predictive analytics.
Teaching Methods: Chalk& Board/ PPT/Video Lectures
Text Books/Suggested References: Text Books: 1. Anasse Bari, Mohamed Chaouchi, Tommy Jung, “Predictive Analytics for Dummies”, Wiley Publisher, 2nd Edition, 2016 2. Bertt Latnz, Machine Learning with R: Expert techniques for predictive modeling to solve all your data analysis problems, Pack Publisher, 2nd Edition, 2015 Reference Books: 1. Aurelian, “Hands-On Machine Learning with Scikit- Learn & Tensorflow”, O’Reilly Publisher, 5th Edition, 2017. 2. Max Kuhn, Kjell Johnson, “Applied Predictive Modeling” Springer, 2013. E BOOKS 1. https://vuquangnguyen2016.files.wordpress.com/2018/03/applied-predictive-modeling-maxkuhn-kjell-johnson_1518.pdf 2. https://www.researchgate.net/publication/329873035_Prediction_Modeling_Methodology 3. https://www.memsql.com/releases/oreilly-predictive-analytics/ MOOC 1. https://www.coursera.org/learn/predictive-modeling-analytics 2. https://www.edx.org/course/predictive-analytics 3. https://www.udemy.com/course/machinelearningandlogisticregression

V Semester [Third Year]
B. Tech in Computer Science and Engineering (Data Science) (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	HS	23BMGMG35001 / 23BMGMG35002	Engineering Economics and Costing/ Optimization Engineering	2	0	0	2	40	60
2	PC	23BCDPC35001	Software Engineering	3	0	0	3	40	60
3	PC	23BCDPC35002	Business Intelligence and Analytics	3	0	0	3	40	60
4	PC	23BCDPC35003	Language Theory and Compiler Design	3	0	0	3	40	60
5	PE	23BCDPE350**	Professional Elective-III (From Bucket)	3	0	0	3	40	60
6	OE	23BCDOE350**	Open Elective-I (From Bucket)	3	0	0	3	40	60
			TOTAL	17	0	0	17	240	360
PRACTICAL/SESSIONAL									
7	PC	23BCDPC35101	Advanced Programming Lab	0	0	2	1	30	20
8	PC	23BCDPC35102	Business Intelligence and Analytics Lab	0	0	2	1	30	20
9	PS T	23BCDPW3510 1	Minor Project -I	0	0	4	2	50	50
10	PS T	23BCDPW3510 2	Summer Internship-II	0	0	1	1	0	50
11	MC	23BMCMC30001	Constitution of India	0	0	0	0	#100	
			TOTAL	0	0	9	5	110	140
			TOTAL	17	0	9	22	350	500

Professional Elective Bucket

Sl. No	Sem	Professional Elective III
1	23BCDPE35011	Tools and Techniques for Data Science
2	23BCDPE35012	R for Data Science

BUCKETS FOR OPEN ELECTIVES

Sl. No	Sem	Open Elective I
1	23BCDOE35011	Web Technology
2	23BCDOE35012	Advanced Java
3	23BCDOE35013	Full Stack Development

Assignment Based Assessment

Subject Code		Name of the Subject								L	T	P	C		
		Language Theory and Compiler Design								3	0	0	3		
Course Educational Objectives															
CEO1	To introduce the major concept areas of language translation and compiler design.														
CEO2	To enrich the knowledge in various phases of compiler and its use.														
CEO3	To provide practical programming skills necessary for constructing a compiler														
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Understand the fundamental concepts and phases involved in the design and development of a compiler.														
CO2	Apply lexical and syntax analysis techniques for language translation using various parsing methods.														
CO3	Design and manage symbol tables and perform intermediate code generation for different programming constructs														
CO4	Implement semantic analysis and syntax-directed translation for type checking and expression evaluation.														
CO5	Develop code generation algorithms and optimize code for efficient memory and execution														
CO6	Use compiler generation tools to construct machine-dependent code and evaluate the effectiveness of various optimization techniques.														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	2	2												
CO2	2	1	2												
CO3	1	1	3												
CO4	3	2	3												
CO5	1	1	3												
CO6	1	1	1												
Avg.	1.5	1.3	2.3												
SYLLABUS															
UNIT:1 (12 Hours)															
Introduction to Compiler Design and Lexical Analysis, Overview of Compilation and its Phases, Lexical Analysis: Role of a Lexical Analyzer, Finite Automata: Non-Deterministic (NFA) and Deterministic (DFA), Regular Expressions, Lexical Grammar, and Design of a Lexical Analyzer, Tools for Lexical Analysis: Lex and Flex.															
UNIT:2 (12 Hours)															
Syntax Analysis: Role of a Parser, Context-Free Grammars, and Languages, Parse Trees and Derivations, Ambiguous Grammar, Top-Down Parsing: Recursive Descent Parsing, LL(1) Grammars, Bottom-Up Parsing: Shift-Reduce Parsing, SLR Parsers, LR(1) Parsers, LALR Parsers, Error Handling, Reporting, and Recovery Techniques															
UNIT:3 (10 Hours)															
Syntax-Directed Translation and Semantic Analysis, Syntax-Directed Definitions (SDD), Inherited and Synthesized Attributes, Dependency Graphs and Evaluation Orders, Semantic Analysis: Semantic Rules and Attribute Grammars, Applications of Syntax-Directed Translation															

UNIT:4**(10 Hours)**

Intermediate Code Generation and Symbol Table Management: Symbol Table: Structure, Attributes, and Scope Management, Intermediate Code Generation: Three Address Code, Quadruples, and Triples, Representations of Syntax Trees and DAG for Expressions, Type Checking, Type Conversions, and Control Flow Statements, Intermediate Code for Procedure Calls and Run-Time Environments

UNIT:5

Code Generation and Optimization, Code Generation: Basic Blocks, Flow Graphs, and Code Generation Techniques, Register Allocation and Assignment Techniques, Code Optimization: Peephole Optimization, Elimination of Local Common Sub-expressions, Flow Control Optimization, Removal of Redundant Code, and Unreachable Code, Introduction to Machine-Dependent Optimization

Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert

Text Books:

1. *Compilers – Principles, Techniques and Tools* Authors: Alfred V. Aho, Monica S. Lam, Ravi Sethi and Jeffrey D. Ullman Publisher: Pearson

Reference Books:

1. *Morden Compiler Design*, D. Galles, Pearson Education.
2. *Compiler Design in C*, Allen I. Holub, PHI

Subject Code		Name of the Subject								L	T	P	C			
		Software Engineering								3	0	0	3			
Course Educational Objectives																
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>																
CO1	Learn various software development process models and their suitability.															
CO2	Able to apply the methods of requirement elicitation.															
CO3	Learn Conventional Software Management and Evolution of Software Economics.															
CO4	Explain how a software process flows and understand Project Organizations and Responsibilities.															
CO5	Apply Agile methodologies and DevOps practices to enhance software development efficiency and collaboration															
CO6	Explore and integrate emerging trends and advanced techniques in software engineering for future-proof software solutions.															
CO-PO & PSO Mapping																
Cos	PROGRAMMEOUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1		3				2				1	1					
CO2		2		3												
CO3	1		1		2											
CO4		1														
Avg.	1	1.5	1	3	2					1	1					
SYLLABUS																
UNIT:1 (10 Hours)																
Introduction to Software Engineering, Software Process, Perspective and Specialized Process Models: The Waterfall Model, Incremental Process Models, The RAD Model, Prototyping Model, V- Model, Spiral Model, Agile and Scrum Model. COCOMO Model																
UNIT:2 (12 Hours)																
Software Requirements: Functional and Non-functional, User requirements, System requirements, Software Requirements Document, Requirement Engineering Process: Feasibility Studies, Requirement elicitation and analysis, Requirement validation, Requirements Management. CASE TOOL: Application on Documentation																
UNIT:3 (12 Hours)																
Design process: Design concepts, Design Model, Design Heuristic Architectural Design: Architectural Mapping using Data Flow Structure Chart design using DFD. Transform Analysis, Transaction Analysis: OOD Modeling Using UML, User Interface Design.																
UNIT:4 (14 Hours)																
Software implementation techniques: coding practices: Refactoring, Code Review, Code Inspection, Driver and Stub Module. Software Testing Fundamentals: Internal and external views of Testing- White Box Testing, Basis path testing, control structure testing. Black Box Testing: Regression Testing, Unit Testing, Integration Testing, Validation Testing, System Testing and Debugging. CASE TOOL: Test-Case Design, Software Reliability Measures and Growth Modeling: SEI CMM: Characteristics of Software Maintenance, Reverse Engineering, Re-engineering.)																

UNIT-V

Emerging Trends and Advanced Topics in Software Engineering, Agile Software Development: Scrum, Kanban, Extreme Programming (XP), and Lean DevOps: Principles, Practices, and Tools , Cloud-Based Software Engineering and Microservices Architecture, Secure Software Development: Secure Coding Practices, Threat Modeling, and Vulnerability Analysis, Future Trends: AI in Software Engineering, Quantum Computing, and Green Software Engineering

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Book:

1. *Fundamentals of Software Engineering*, Rajib Mall, PHI, 2014.
2. *Walker Rayce : Software Project Management*, Pearson Education, 2005

Reference Book

1. *Software Engineering, A Practitioner's Approach*, Roger S. Pressman, TMG Hill.
2. *Software Engineering*, I. Sommerville, 9th Ed. , Pearson Education
3. *Richard H.Thayer: Software Engineering Project Management*, IEEE Computer Society, 1997
4. *ShereK.D. : Software Engineering and Management*, Prentice Hall, 1988

Subject Code	Business Intelligence and Data Analytics	L	T	P	C									
		4		1										
Course Educational Objectives														
CEO 01	Explain the Business Intelligence, Analytics and Decision Support system													
CEO 02	List the technologies for Decision making, Automated decision systems													
CEO 03	Explain sentiment analysis techniques													
CEO 04	Illustrate Multi-criteria Decision making systems, predictive modelling techniques													
Course Outcomes: : Upon successful completion of this course, students should be able to:														
CO 01	Analyze Business Intelligence, Analytics and Decision Support and Business Intelligence Applications													
CO 02	Explain the technologies for Decision making													
CO 03	Apply Mathematical modelling techniques and Data preparation Techniques													
CO 04	Apply various Modelling Techniques in Business Intelligence using ML techniques													
CO 05														
CO 06														
CO-PO & PSO Mapping (Articulation Matrix):														
COs	PROGRAMME OUTCOMES												PSOs	
	01	02	03	04	05	06	07	08	09	10	11	12	01	02
01	2	-	-	1	-	-	-	-	-	-	-	1	1	-
02	2	2	2	1	-	-	-	-	1	-	-	1	1	1
03	2	2	2	1	-	-	-	-	1	-	-	1	1	1
04	2	2	2	1	-	-	-	-	1	-	-	1	1	1
05														
06														
Avg.	2.0	2.0	2.0	1.0	-	-	-	-	1.0	-	-	1.0	1.0	1.0
Syllabus														
UNIT I													12 Hours	
Introduction Business intelligence: Introduction, History BI, Types of BI Effective and timely decisions, Data, information and knowledge, The role of mathematical models, Business intelligence architectures, Ethics and business intelligence Decision support systems: Definition of system, Representation of the decision-making process, Evolution of information systems, Definition of decision support system, Development of a decision support system														
UNIT II													12 Hours	
Mathematical models for decision making: Structure of mathematical models, Development of a model, Classes of models Data mining: Definition of data mining, Representation of input data , Data mining process, Analysis methodologies Data preparation: Data validation, Data transformation, Data reduction														

UNIT III		12 Hours
Classification: Classification problems, Evaluation of classification models, Bayesian methods, Logistic regression, Neural networks, Support vector machines Clustering: Clustering methods, Partition methods, Hierarchical methods, Evaluation of clustering models		
UNIT IV		09 Hours
Business intelligence applications: Marketing models: Relational marketing, Sales force management, Logistic and production models: Supply chain optimization, Optimization models for logistics planning, Revenue management systems.		
UNIT V		09 Hours
Data envelopment analysis: Efficiency measures, Efficient frontier, The CCR model, Identification of good operating practices. Case Study: Case Studies on BI using Microsoft Power BI and Tableau		
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs		
Text Books: 1. Business Intelligence: Data Mining and Optimization for Decision Making, Carlo Vercellis 3. Decision support and Business Intelligence Systems, Efraim Turban, Ramesh Sharda, Dursun Delen 4. Fundamental of Business Intelligence, Grossmann W, Rinderle-Ma		

Subject Code		SUBJECT NAME										L	T	P	C
		R for Data Science											4	0	3
Course Educational Objectives															
CEO 01	To Learn the basic concepts of R.														
CEO 02	To know various input and output functions with readr, dplyr packages.														
CEO 03	To understand various data interfaces and working with data sets.														
CEO 04	To know data collection, preparation, analysis, modeling and visualization of data, covering both conceptual and practical issues.														
Course Outcomes: : Upon successful completion of this course, students should be able to:															
CO 01	Understand and use the R programming environment, basic syntax, data types, and structures for data manipulation.														
CO 02	Use R functions for input/output operations, data manipulation, and control structures to manage data efficiently.														
CO 03	Import and manage datasets from various file formats using R packages and interfaces.														
CO 04	Create data visualizations and perform statistical analyses using R to interpret and present data insights.														
CO 05	Apply various data manipulation techniques in R, including data cleaning, transformation, and summarization, to prepare datasets for analysis.														
CO 06	Evaluate the effectiveness of different visualization techniques in R for accurately communicating data insights and make informed decisions on selecting appropriate visualizations for different types of data.														
CO-PO & PSO Mapping (Articulation Matrix):															
COs	PROGRAMME OUTCOMES												PSOs		
	01	02	03	04	05	06	07	08	09	10	11	12	01	02	
01	3	2	2	1	2	1						1			
02	3	2	2	2	1										
03	2	3	2	2	2										
04	3	3	1	3	2										
05	3	3	1												
06	3	3	1												
Avg.	2.75	2.5	1.75	2	1.75	1						1			
Justification of the Mapping:															
CO	PO	Level	Justification												
01	01	3	Understanding R basic in Engineering Knowledge.												
01	02	2	Data Manipulation is a kind of Problem Analysis (Medium)												
02	01	3	Problem Analysis using control structures and manage data.												
02	02	2	Using the dataset loaded we try to do the analysis of Problems.												
03	02	3	Managing and understanding dataset is an Engineering Knowledge.												
04	01	3	Visualization is a part of Engineering Knowledge.												
Syllabus															
UNIT I		Introduction										14 Hours			
History and overview of R Uses, getting started with the R interface, R Environment Setup, Basic Syntax, R-Objects/Data Types, Variables, Operators, Creating Vectors, Mixing Objects, Explicit Coercion, Matrices, Lists, Factors, Missing Values, Data Frames, Names, Decision Making, Loops, Functions, Strings, Arrays, Factors, Packages, Data Reshaping, Dates and Time.															

R-Keywords, Limitations of R.		
UNIT II	Working With R Functions	14 Hours
Taking Input, Printing Output, Print the Argument Functions in R Programming, Getting Data In and Out of R, Using the readr Package, Managing Data Frames with the dplyr package, Interfaces to the Outside World, Subsetting R Objects, Control Structures, Looping.		
UNIT III	Packages, Data Interface and Datasets	10 Hours
Tidyverse, CSV Files, Excel Files, Binary Files, XML Files, JSON Files, Web Data, Database Working with datasets: Airquality, JohnsonJohnson, mtcars, Worldphones, Quakes		
UNIT IV	Data Visualisation and Statistical Analysis	12 Hours
The Layered Grammar of Graphics; R Charts & Graphs: Pie Charts, Bar Charts, Boxplots, Histograms, Line Graphs, Scatterplots Statistical Analysis: Mean, Median & Mode, Linear Regression, Multiple Regression,		
UNIT V		12 Hours
Logistic Regression, Normal Distribution, Binomial Distribution, Poisson Regression, Analysis of Covariance, Time Series Analysis, Nonlinear Least Square, Decision Tree, Random Forest, Survival Analysis, Chi Square Tests		
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs		
Text Books:		
1. Roger D. Peng, "R Programming for Data Science" Leanpub.		
2. Hadley Wickham and Garrett Grolemund - "R for Data Science" O'Reilly Media, Inc.		
Reference Book:		
1. Ross Ihaka and Robert Gentleman, "R Programming" Tutorials Point (I) Pvt. Ltd.		
2. Garrett Grolemund, "Hands-On Programming with R" ,O'Reilly Media, Inc.		

	Advanced Java Programming										4	0	0	4
Course Educational Objectives														
CEO1	Objective of this course is to provide the ability to design console based, GUI based and web based applications. Students will also be able to understand integrated development environment to create, debug and run multi-tier and enterprise-level applications.													
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>														
CO1	design and develop data driven web applications using JDBC, servlet and JSP API.													
CO2	develop MVC based web applications using SpringBoot.													
CO3	understand significance of Object Relational Mapping (ORM) in Java context.													
CO4	construct ORM-based web applications using Hibernate.													
CO5	Develop proficiency in advanced Java programming concepts.													
CO6	Utilize Java's networking and I/O capabilities to build robust and scalable													
CO-PO & PSO Mapping (Articulation Matrix):														
Cos	PROGRAMME OUTCOMES												PSOs	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	1	1	3				3						
CO2		3	2					3						
CO3	2													
CO4		2	2											
CO 6			2											
CO 7			2											
Avg.	2.5	2	1.8					3						
SYLLABUS														
UNIT:1 (12 Hours)														
Basics of JDBC: Introduction to JDBC, Need of JDBC, JDBC Drivers (4 types), Architecture of JDBC, Components of JDBC(Classess and Interfaces).														
Programming with JDBC: Creating a DATABASE (ORACLE), First Program to connect to the DATABASE created, Loading the Driver, Establishing the Connection, Creating Statements (Statement/ Prepared Statement/ Callable Statement), Executing a SQL Query, Different types of SQL Queries.														
UNIT:2 (12 Hours)														
JDBC Program to Retrieve Data from DATABASE: Introduction to Result Set, Result Set with Statement Interface, Result Set with Prepared Statement Interface, Updating and deleting data in database, Result Set Meta data, Executing Stored Procedures Using Callable Statement.														
Web Architecture: Introduction to HTML, Describe the various HTML tags and their usage, Layout Tags, Semantic Tags, HTML table, forms, style, div.														
UNIT:3 (10 Hours)														
Introduction to Servlets: What is Servlet, Advantage of Servlet Over Applets and CGI, Strengths of Servlet, Architecture of Web Application, Web Servers and its Containers, Role of servlet in Web application development, Understanding servlet-api, Understanding HTTP protocol and communication between HTML-SERVLET.														
Getting Deep into Servlets: Types of Servlet, Difference between HttpServlet and GenericServlet, Life cycle of Servlets and different life cycle methods, Difference between doGet() and doPost(), Servlet Generating HTML output, Collecting Client submitted data in a Servlet.														
UNIT:4 (14 Hours)														
Servlet Communications: Servlet to DBMS communication using type-4 connection, Servlet to DBMS														

communication using JDBC connection pooling, Servlet communication with other servlets (Servlet Chaining), Servlet communication with JSP or HTML page (sendRedirect(), Difference between sendRedirect() and RequestDispatcher forward(), Understanding ServletConfig. Conclusion to Servlets: Servlet Filters and Wrappers, Servlet Listeners, Session Tracking, Cookies, HttpSession, HTML hidden form field element, URL rewriting, Annotation-based servlet programs..

UNIT:5

(14 Hours)

Java Server Pages (JSP): Introduction to JSP, Scope of JSP, Anatomy of a JSP program, Execution of a JSP program, Significance of JSP Engine, Built-in objects of JSP, Significance of JSP Elements, Scripting Elements, Scriptlets, Declaration, Expression, Directives and Action Elements, Page Directive, Include Directive, Taglib Directive, Forward action element, Include, Param, useBean with introduction to beans, setProperty, getProperty. Hibernate: Introduction to Hibernate, Architecture of Hibernate, Hibernate O/R Mapping, Configuring Hibernate development environment, Implementing Hibernate O/R Mapping, Introduction to Hibernate Query Language (HQL). Spring and Spring Boot: Spring Basics, Spring IOC, Spring MVC, Spring MVC with Hibernate, Introduction to Spring Boot, Spring Boot Injection, Dependency, Autowired, Spring Boot JPA, Spring Boot Webapp MVC.

Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs

Text Books:

1. Advanced Java Programming, Uttam K. Roy, Oxford University Press.

Reference Book:

1. Black book, Kogent Learning Solution Inc.
2. Java 2: The Complete Reference by Herbert Schildt, Fifth Edition Paperback

VI Semester [Third Year]
B. Tech in Computer Science and Engineering (Data Science) (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	HS	23BMGMG35001 / 23BMGMG35002	Engineering Economics and Costing/ Optimization Engineering	2	0	0	2	30	70
2	PC	23BCDPC36001	Artificial Intelligence	3	0	0	3	30	70
3	PC	23BCDPC36002	Microprocessors and Microcontroller	3	0	0	3	30	70
4	PC	23BCDPC36003	Introduction to Internet of Things	3	0	0	3	30	70
5	PE	23BCDPE360**	Professional Elective-IV (From Bucket)	3	0	0	3	30	70
6	OE	23BCDOE360**	Open Elective-II (From Bucket)	3	0	0	3	30	70
			TOTAL	17	0	0	17	180	420
PRACTICAL/SESSIONAL									
7	PC	23BCDPC36103	Internet of Things Lab	0	0	2	1	30	20
8	PC	23BCDPC36101	Machine Learning Lab	0	0	2	1	30	20
9	PST	23BCDPW36101	Minor Project Work-II	0	0	4	2	50	50
10	MC	23BMCMC30002	Essence of Indian Knowledge Tradition	0	0	0	0	#100	
			TOTAL	0	0	8	4	110	90
			TOTAL	17	0	8	21	290	510

Professional Elective Bucket

Sl. No.	Sem	Professional Elective IV
1	23BCDPE36011	Machine Learning
2	23BCDPE36012	Data Modelling & Simulation

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Sem	Open Elective II
1	23BCDOE36011	Android Application Development using Kotlin
2	23BCDOE36012	Mobile Computing
3	23BCDOE36013	IOS Development

Assignment Based Assessment

Subject Code		Name of the Subject										L	T	P	C	
		Machine Learning										3	0	0	3	
Course Educational Objectives																
CEO1	Teach fundamentals of machine learning and regression algorithms															
CEO2	Teach the role of machine learning classification in business decision-making															
CEO3	Teach Model building and validation using various techniques															
Course Outcomes: Upon successful completion of this course, students should be able to:																
CO1	Understanding the regression and interpreting the gradient descent way															
CO2	To Implement the classification algorithm															
CO3	Analyze the algorithm and perform the robustness of the model by combining the different classifiers															
CO4	Develop intelligent decision support systems															
CO 5	Analyze different types of Machine Learning architectures (e.g., supervised, unsupervised, reinforcement) and differentiate between various algorithms used for classification, regression, and clustering based on their advantages and limitations.															
CO 6	Design and implement machine learning models using Python, including the selection and tuning of hyperparameters, to solve complex problems and optimize model performance across different learning paradigms.															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	3	3														
CO2	3	3	2													
CO3	3	3	3													
CO4	3	2	3	2	2											
CO 5	3	2	3	2												
CO 6	3	2	3	2												
Avg.	3	2.5	2.8	2	2											
SYLLABUS																
UNIT:1 (12 Hours)																
Introduction to Machine Learning :																
Introduction, Components of Learning, Learning Models, Types of Learning, Supervised, Unsupervised, Reinforcement, Perspectives and Issues, Understand your Data and Prepare Your Data For Analysis. Bias, variance Introduction of Regression and classification, Linear Methods for Regression, Least squares method, Linear algebra solution to the least square problem, Multiple linear regression, Regularization, Ridge Regression, Lasso regression, Interpreting the OLS Regression Results, Neural Network Regression, Linear Regression Model using Gradient Descent algorithm. Error estimation of regression algorithms																
UNIT:2 (12 Hours)																
Machine learning classification approaches Logistic Regression, Naive Bayes, K-Nearest Neighbors, and Support Vector Machines. Cost function of logistic regression, Gradient descent in logistic regression. Multiple linear regression with gradient descent, stochastic gradient descent, mini-batch gradient steepest descent																

UNIT:3	(12 Hours)
Ensemble and Probabilistic Learning Model Combination Schemes, Voting, Bagging: Random Forest Trees, Boosting: Adaboost, Stacking. Gaussian mixture models - The Expectation-Maximization (EM) Algorithm Inferential Statistics and Sensitivity Analysis	
UNIT:4	(12 Hours)
Estimation Strategies, Accuracy Estimation, Performance Estimation of Classifier, Confusion Matrix, Performance Evaluation Metrics, ROC Curve, Hypothesis testing. Parametric and non parametric test)	
UNIT:5	(12 Hours)
Neural Networks & Deep Learning Introduction of Artificial Neural Network (ANN), Perceptron (Single Artificial Neuron), Multilayer Perceptrons (Feedforward Neural Network), Perceptron Training algorithm for single /multiple output classes, Convolution Neural Networks (CNN), Recurrent Neural Networks (RNN): Long Short-Term Memory (LSTM), LSM, Generative Adversal Networks (GAN).	
Teaching Methods: Chalk & Board/ PPT/Video Lectures	
Text Books: 1. <i>Probability & Statistics for Engineers & Scientists (9th Edn.)</i>, Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, Prentice Hall Inc. 2. <i>The Elements of Statistical Learning, Data Mining, Inference, and Prediction (2nd Edn.)</i>, Trevor Hastie Robert Tibshirani Jerome Friedman, Springer, 2014 3. <i>An Introduction to Statistical Learning: with Applications in R</i>, G James, D. Witten, T Hastie, and R. Tibshirani, Springer, 2013 4. <i>Data Mining and Analysis</i>, Mohammed J. Zaki, Wagner Meira, Cambridge, 2012	
Reference Books: 1. <i>Software for Data Analysis: Programming with R (Statistics and Computing)</i>, John M. Chambers, Springer 2. <i>Data Mining and Business Analytics with R</i>, Johannes Ledolter, John Wiley & Sons, Inc. 2013	

Subject Code		Name of the Subject										L	T	P	C	
		Android App Development Using Kotlin										4	0	1	4	
Course Educational Objectives																
CE01	This course covers the fundamentals of Kotlin Programming and Android programming using the Android SDK. Topics discussed in this course include: fundamental concepts in Android programming - activities and intents, designing user interface using views, data persistence, content providers, messaging and networking, location-based services, and developing android services.															
Course Outcomes: Upon successful completion of this course, students should be able to:																
CO1	demonstrate their understanding of Kotlin Programming.															
CO2	demonstrate their understanding of the fundamentals of Android operating systems, development trends and Android platform.															
CO3	Design kotlin based stand android application.															
CO4	Develop android application using API and database.															
CO5	Implement different types of Menus															
CO6	Usage of Data Storage and SOLLite															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES															
	1	2	3	4	5	6	7	8	9	10	11	12				
CO1	3	1			1							1				
CO2	2	3	1	3	2							1				
CO3	2	3	3	3	2				1		1	2				
CO4	2	3	3	3	2				1		1	2				
Avg.	2.25	2.5	2.3	3	1.75				1		1	1.5				
SYLLABUS																
UNIT:1 (8 Hours)																
Introduction to Kotlin: Kotlin Architecture, Basic syntax, Kotlin Comments, Kotlin Keywords, Defining variable, Use of variables, Data types- (Booleans, Strings, Arrays, Ranges), Operators, Control Flow, if... else expression, for loops, While loops, Break and Continue, functions in Kotlin, user defining functions. Kotlin Collections (Lists, Sets, Maps), Kotlin objects and Classes, Constructors, Inheritance, Interface, Visibility Control, Extension, Data Classes, Sealed Class, Generics, Delegation, Destruction, Exception Handling.																
UNIT:2 (15 Hours)																
Introduction to Android: Introduction to Android Operating System, Android Architecture, Core Building Blocks, Android Emulator, Application Structure, Setting up development environment, Dalvik Virtual Machine & .apk file extension, AndroidManifest.xml, uses-permission & uses-sdk, Resources & R.java o Assets, Layouts & Drawable Resources, Fundamentals: Basic Building blocks - Activities, Services, Broadcast, Receivers & Content providers, UI Components - Views & notifications, Components for communication -Intents & Intent Filters Android API levels (versions & version names), User Interface Architecture: Application context, intents, Activity life cycle, Services, Android widgets, UI widgets, Working with Buttons, Toast, Custom Toast, Toggle Button, Check Button, Custom Check Button, Radio Button, Dynamic Radio button, Alert Dialog, Android Views: Text view, Autocomplete Text																

view, List View, Custom List View, Rating Bar, WebView Seek Bar, Date Picker, Time Picker, Analog and Digital, Progress Bar, Android Layouts: Linear Layout, Relative Layout, Table Layout, Table Layout with Frame Layout, Search View, Search View on Toolbar, Edit Text with Text Watcher.	
UNIT:3	(10 Hours)
Switchers: Switch, Dynamic Switch, Text Switcher, Dynamic Text Switcher, Image Switcher, Dynamic Image Switcher. Spinner: Spinner, Dynamic Spinner, Dynamic TextClock, Chronometer, Dynamic Chronometer, Notifications, Android Toast, Dynamic RadioGroup, Android Slide Up/Down in Kotlin, Scroller: Vertical Scroll, View Horizontal Scroll View, Image Switcher, Image Slider, View Stub, XML Parsing using DOM Parser, XML Parsing using SAX Parser, Kotlin Android XMLPullParser	
UNIT:4	(10 Hours)
Menu : Option menu, Context menu, Sub menu, menu from xml, menu via code, Examples. Navigation: Create a fragment, Define navigation paths, Start an external activity, Kotlin Regex: Regular Expressions, Kotlin Annotations, Adapters, Android RecyclerView, Testing Android applications, Publishing Android application, Using Android preferences, Managing Application resources in a hierarchy, working with different types of resources	
UNIT:5	
Data Storage: Shared Preferences, Android File System, Internal storage, External storage, SQLite Firebase. Common Android APIs: Using Android Data and Storage APIs etc.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books:	
1. Bruce Eckel and Svetlana Isakova (2021), “Atomic Kotlin”, Mindview LLC, Crested Butte, CO 2. WeiMeng Lee (2012), “Beginning Android Application Development”, Wrox Publications (John Wiley, New York) (For 1 to 5 units).	
Reference Books:	
1. Ed Burnette (2010), “Hello Android: Introducing Google's Mobile Development Platform”, The Pragmatic Publishers, 3rd edition, North Carolina USA 2. Reto Meier (2012), “Professional Android 4 Application Development”, Wrox Publications (John Wiley, New York). 3. ZigurdMednieks, Laird Dornin, Blake Meike G, Masumi Nakamura (2011), “Programming Android: Java Programming for the New Generation of Mobile Devices”, OReilly Media, USA	
Contents:	
1. https://developer.android.com/kotlin 2. https://developer.android.com/studio	

Subject Code		Name of the Subject								L	T	P	C		
		Microprocessors and Microcontrollers								3	0	2	4		
Course Educational Objectives															
CEO1	To Develop assembly language programs and basic concepts of the microprocessor and microcontroller														
CEO2	To provide solid foundation on interfacing the external devices to the microprocessor & microcontroller according to the user requirements in order to create novel products and solutions for the real time problems														
CEO3	To Familiar and Design of any type of embedded systems related to industrial and real time applications by knowing the concepts of Microprocessor and Microcontrollers.														
CEO4	To assist the students with an academic environment aware of excellence guidelines and lifelong learning needed for a successful professional carrier in the field embedded systems.														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	Understand the architecture of 8085, 8086 & 8051														
CO2	Understand and analyze different data transfer schemes and its applications														
CO3	To design circuits and Implement programs using microcontrollers.														
CO4	Understand the need & use of Interrupt structure in 8085, 8086 & 8051.														
CO5	Analyze and troubleshoot microprocessor and microcontroller systems.														
CO6	Apply microcontrollers in real-world embedded system applications.														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	1													
CO2	3	2													
CO3	2	2	1												
CO4	3	2													
CO5	1	1													
CO6	2	2													
Avg.	2.3	1.6	1												
SYLLABUS															
UNIT:I (10 Hours)															
Introduction to 8 bit Microprocessors															
Introduction to 8085 microprocessor, Architecture, Signal Descriptions ,Buses-Address bus ,data bus and control Bus ,Instruction format ,Instruction sets ,addressing Modes, Assembly Language Programming ,Timing diagram, stack and sub routine, Data Transfer Schemes ,Memory Interfacing and 8085 interrupts.															
UNIT:2 (12 Hours)															
Advanced Microprocessor															
Introduction to 8086 microprocessors, 8086 Architecture, Register Organization, signal descriptions, Memory Segmentation. Physical memory organization. Addressing Modes, instruction Set Minimum and Maximum mode operation, Bus Cycle of minimum mode and maximum mode. Interrupts of 8086, Memory interfacing & Assembly Language Program.															
UNIT:3 Peripheral Devices (10 Hours)															

Programmable Peripheral Interface (8255), Programmable Interval Timer (8254) Programmable Interrupt Controller (8259A) - Programmable DMA Controller(8257),Programmable Communication Interface (8251A) – Programmable Keyboard and Display Controller (8279).	
UNIT:4	(12 Hours)
8051 Microcontroller Overview of 8051 microcontroller. Architecture. I/O Ports. Memory organization, Addressing modes, data transfer instructions, Logical instructions, Arithmetic instructions, Branching (Jump & Call) instructions, Bit addressable instructions and special instructions, Interrupts and interrupt handler sub routines (Interrupt Service Routines).Assembly language program.	
UNIT-5	(10 Hours)
Advanced Microcontrollers and Embedded Systems: Introduction to advanced microcontrollers: ARM, AVR, and PIC, Comparison of different microcontroller architectures, Embedded system design: Hardware/software co-design, RTOS basics, Application of microcontrollers in embedded systems, Case studies: Real-time applications using microcontrollers	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. <i>Microprocessor Architecture, Programming and application with 8085</i> , R.S. Gaonkar, PRI Penram International publishing PVT. Ltd., 5thEdition 2. <i>Advanced Microprocessors and Peripherals</i> - A. K. Ray and K.M. Bhurchandani, TMH, 2nd edition 2006. 3. <i>The 8051 Microcontroller and Embedded Systems</i> , Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D.M C Kinlay, Pearson Education, Second Edition, 2008.	
Reference Books: 1. <i>Microprocessors and Microcontrollers Architecture, programming and system design using 8085, 8086, 8051 and 8096</i> , Krishna Kant, PHI Publication, 2007. 2. <i>Microprocessors and Interfacing, Programming and Hardware</i> , Douglas V Hall, TMH Publication, 2006. 3. <i>Microprocessors and Interfacing</i> , N. Senthil Kumar, M. Saravanan, S. Jeevananthan and S.K. Shah, Oxford University Press.	

Subject Code		Name of the Subject								L	T	P	C			
		Introduction to Internet of Things								3	0	2	4			
Course Educational Objectives																
CEO1	The objective of this course is to impart necessary and practical knowledge of components of Internet of Things and develop skills required to build real-life IoT based projects.															
Course Outcomes: Upon successful completion of this course, students should be able to:																
CO1	Understand internet of Things and its hardware and software components.															
CO2	Interface I/O devices, sensors & communication modules.															
CO3	Remotely monitor data and control devices.															
CO4	Develop real life IoT based projects.															
CO5																
CO6																
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES															
	1	2	3	4	5	6	7	8	9	10	11	12				
CO1	3	1			1							1				
CO2	2	3	1	3	2							1				
CO3	2	3	3	3	2				1		1	2				
CO4	2	3	3	3	2				1		1	2				
CO5	2	2	3													
CO6	1	1	3													
Avg.	2	2.1	2.6	3	1.75				1		1	1.5				
SYLLABUS																
UNIT:1 (8 Hours)																
Introduction to IoT: - Architectural Overview, Design principles and needed capabilities, IoT Applications, Sensing, Actuation, Basics of Networking, M2M and IoT Technology Fundamentals- Devices and gateways, Data management, Business processes in IoT, Everything as a Service(XaaS), Role of Cloud in IoT, Security aspects in IoT.																
Domain-Specific IOTs: Home Automation, Cities, Environment, Energy, Retail, Logistics, Agriculture, Industry, Health & Life Style.																
UNIT:2 (9 Hours)																
Elements of IoT: - Hardware Components- Computing (Arduino, Raspberry Pi), Communication, Sensing, Actuation, I/O interfaces.																
Software Components- Programming API's (using Python/Node.js/Arduino) for CommunicationProtocols-MQTT, ZigBee, Bluetooth, CoAP, UDP, TCP.																
UNIT:3 (18 Hours)																
IoT Application Development: - Solution framework for IoT applications- Implementation of Device integration, Data acquisition and integration, Device data storage- Unstructured data storage on cloud/local server, Authentication, authorization of devices.																
UNIT:4 (10 Hours)																
IoT Case Studies: - IoT case studies and mini projects based on Industrial automation, Transportation, Agriculture, Healthcare, Home Automation.																
UNIT-5 (10 Hours)																
IoT Data Management and Analytics: Data generation, acquisition, and processing in IoT, IoT																

data storage and management: Cloud storage, edge computing, and big data, Data analytics for IoT: Machine learning, predictive analytics, and real-time processing, Integration with cloud platforms: AWS IoT, Azure IoT, Google Cloud Io, Visualization and dashboards for IoT data.

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Books:

1. VijayMadiseti, Arshdeep Bahga, "Internet of Things A Hands-On-Approach", 2014, ISBN: 978 0996025515.
2. Dr. SRN Reddy, Rachit Thukral and Manasi Mishra, "Introduction to Internet of Things: A practical Approach", ETI Labs.
3. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press.
4. Jeeva Jose, "Internet of Things", Khanna Publishing House, Delhi.

Reference Books:

5. AdrianMcEwen, "Designing the Internet of Things", Wiley Publishers, 2013, ISBN: 978-1-118-43062-0 .
6. Daniel Kellmerit, "The Silent Intelligence: The Internet of Things". 2013, ISBN: 0989973700
7. Adrian McEwen, "Designing the Internet of Things", Wiley.
Raj Kamal, "Internet of Things: Architecture and Design", McGraw Hill.
8. Cuno Pfister, "Getting Started with the Internet of Things", O Reilly Media.

eb Books:

9. Designing for the Internet of Things, O'Reilly, February 2015.
10. Getting Started with the Internet of Things, O'Reilly 4th edition, October, 2015.

Subject Code		Name of the Subject								L	T	P	C			
		Artificial Intelligence								3	0	2	4			
Course Educational Objectives																
CEO1	The objective of this course is to impart necessary and practical knowledge of artificial intelligence and develop skills required to build real-life AI based projects.															
Course Outcomes: Upon successful completion of this course, students should be able to:																
CO1	Understand the various characteristics of a problem solving agent															
CO2	Learn about the different strategies involved in problem solving															
CO3	Learn about solving problems with various constraints															
CO4	Apply A.I to various applications like expert systems etc.															
CO5	Understand the different models of learning															
CO6	Design applications like expert systems and chat-bot															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES															
	1	2	3	4	5	6	7	8	9	10	11	12				
CO1	3	1			1							1				
CO2	2	3	1	3	2							1				
CO3	2	3	3	3	2				1		1	2				
CO4	2	3	3	3	2				1		1	2				
CO5	2	2	3													
CO6	1	1	3													
Avg.	2	2.1	2.6	3	1.75				1		1	1.5				
SYLLABUS																
UNIT:1 (8 Hours)																
Problems of AI, AI technique, Tic – Tac – Toe problem. Intelligent Agents, Agents & environment, nature of environment, structure of agents, goal-based agents, utility-based agents, learning agents. Defining the problem as state space search, production system, problem characteristics, and issues in the design of search programs																
UNIT:2 (9 Hours)																
Problem solving agents, searching for solutions; uniform search strategies: breadth first search, depth first search, depth limited search, bidirectional search, comparing uniform search strategies. Heuristic search strategies Greedy best -first search, A* search, AO* search, memory bounded heuristic search: local search algorithms & optimization problems: Hill climbing search, simulated annealing search, local beam search.																
UNIT:3 (18 Hours)																
Local search for constraint satisfaction problems. Adversarial search, Games, optimal decisions & strategies in games, the minimax search procedure, alpha-beta pruning, additional refinements, iterative deepening.																
UNIT:4 (10 Hours)																
Statistical Reasoning: Probability and Bays’ Theorem, Certainty Factors and Rule-Base Systems, Bayesian Networks, Dempster-Shafer Theory, Fuzzy Logic. AI for knowledge representation, rule-based knowledge representation, procedural and declarative knowledge, Logic programming, Forward and backward reasoning.																
UNIT-5 (10 Hours)																
Introduction to Machine Learning: Exploring sub-discipline of AI: Machine Learning,																

Supervised learning, Unsupervised learning, Reinforcement learning, Classification problems, Regression problems, Clustering problems, Introduction to neural networks and deep learning.

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Books:

1. *S. Russell and P. Norvig, "Artificial Intelligence: A Modern Approach", Prentice Hall, Third Edition, 2015.*
2. *Nils J. Nilsson, "Artificial Intelligence: A New Synthesis", 1st Edition, Morgan-Kaufmann, 1998.*
- Technologies, Platforms, and Use Cases", CRC Press.
2. *Jeeva Jose, "Internet of Things", Khanna Publishing House, Delhi.*

Reference Books:

3. *Elaine Rich, Kevin Knight, & Shivashankar B Nair, "Artificial Intelligence", McGraw Hill, 3rd ed.,2017.*
4. *Patterson, "Introduction to Artificial Intelligence & Expert Systems", Pearson, 1st ed. 2015.*
5. *. Saroj Kaushik, "Logic & Prolog Programming", New Age International, 1st edition, 2002.*
6. *Joseph C. Giarratano, Gary D. Riley, "Expert Systems: Principles and Programming", 4th Edition, 2007.*

eB Books:

7. *Designing for the Internet of Things, O'Reilly, February 2015.*
8. *Getting Started with the Internet of Things, O'Reilly 4th edition, October, 2015.*

VII Semester [Fourth Year]
B. Tech in Computer Science and Engineering (Data Science) (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	PC	23BCDPC47001	Computer Graphics	3	0	0	3	40	60
2	PC	23BCDPC47002	Data Communication	3	0	0	3	40	60
3	PE	23BCDPE470**	Professional Elective-V (From Bucket)	3	0	0	3	40	60
4	HS	23BHSHS47001	Entrepreneurship Development	3	0	0	3	40	60
5	OE	23BCDOE470**	Open Elective-III (From Bucket)	3	0	0	3	40	60
			TOTAL	15	0	0	15	200	300
PRACTICAL/SESSIONAL									
5	PC	23BCDPC47102	Data Communication Lab	0	0	2	1	70	30
5	PST	23BCDPW47101	Major Project Work-I	0	0	6	3	50	50
6	PST	23BCDPW47102	Summer Industry Internship-III	0	0	0	1	0	50
			TOTAL	0	0	8	5	120	80
			TOTAL	15	0	8	20	320	380

BUCKETS FOR PROFESSIONAL ELECTIVES

Sl. No.	Semester	Professional Elective V
1	23BCDPE47011	Deep Learning
2	23BCDPE47012	Information Extraction and Retrieval

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Semester	Open Elective III
1	23BCDOE47011	Computer Vision
2	23BCDOE47012	Software Project Management

Subject Code	Subject Code										L	T	P	C	
	Computer Graphics										3	0	0	3	
Course Educational Objectives															
CEO1	Discuss various algorithms for scan conversion and filling of basic objects and their comparative analysis.														
CEO2	Introduce the use of geometric transformations on graphics objects and their application in composite form														
CEO3	Impart frame extraction with different clipping algorithms and transformation to a graphics display device.														
CEO4	Introduce projections and visible surface detection techniques for display of 3D scene on 2D screen and rendering of projected objects to naturalize the scene in 2D view.														
Course Outcomes: : Upon successful completion of this course, students should be able to:															
CO1	To <i>explain</i> the structure of modern computer graphics systems and <i>implement</i> its primitives														
CO2	To <i>design, develop</i> and <i>model</i> key algorithms for modeling and .														
CO3	Apply Graphics in greater depth to more complex courses like Image Processing, Virtual, Augmented Reality, etc														
CO4	To <i>visualize</i> surface detection and Virtual reality for a better visual effect.														
CO5	Desing Illumination models and implement them.														
CO6	Implement Animations in real time applications														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	2													
CO2	2	1													
CO3	1	2	2												
CO4	1	3	2												
CO5	1	2													
CO6	2	1													
Avg.	1.3	1.8	2												
SYLLABUS															
UNIT:1 (10 Hours)															
Overview of Graphics System: Video Display Units, Raster-Scan and Random Scan Systems, Graphics Input and Output Devices, monitor and work station. Graphics output Primitives: point and line, drawing Algorithms: DDA and Bresenham's, Circle drawing Algorithms: Midpoint Circle and Bresenham's filled area primitives.															
UNIT:2 (12 Hours)															
Two Dimensional Geometric Transformations: Translation, rotation, Scaling, Reflection, Shear, Matrix Representation, Composite Transformations, Transformation between coordinate systems. Window-to- View port Coordinate Transformation. Line Clipping (Cohen-Sutherland Algorithm) and Polygon Clipping (Sutherland Hodgeman Algorithm).															
Three Dimensional Geometric and Modeling Transformations: Translation Rotation, Scaling, Reflections, shear, Composite Transformation. Projections: Parallel Projection and Perspective Projection.															

UNIT:3	(10 Hours)
Object representation: Spline Representation, Bezier Curves and B-Spline Curves. Fractal Geometry: Fractal Classification and Fractal Dimension. Visible Surface Detection Methods: Back-face Detection, Depth Buffer, A- Buffer, Scan- line Algorithm and Painters Algorithm.	
UNIT:4	(12 Hours)
Light pattern and Illumination Models: Aliasing and Anti aliasing, Half toning, Thresholding and Dithering, Basic Models, Displaying Light Intensities. Surface Rendering Methods: Polygon Rendering Methods: Gouraud Shading and Phong Shading.	
UNIT-5	[10 Hours]
Computer Animation: Types of Animation, Key frame Vs. Procedural Animation, methods of controlling Animation, Morphing. Virtual Reality: Types of Virtual reality systems, Input and Output Virtual Reality devices	
Teaching Methods: Chalk& Board/ PPT	
Text Books: 1. <i>Computer Graphics with Virtual Reality System, Rajesh K.Maurya, WileyDreamtech.</i> 2. <i>Computer Graphics, D. Hearn and M.P. Baker (C Version), Pearson Education</i>	
Reference Book: 1. <i>Computer Graphics Principle and Practice , J.D. Foley, A.Dam, S.K. Feiner, Addison, Wesley</i> 2. <i>Procedural Elements of Computer Graphics- David Rogers (TMH)</i> 3. <i>Computer Graphics: Algorithms and Implementations – D.P Mukherjee &Debasish Jana (PHI)</i> 4. <i>Introduction to Computer Graphics & Multimedia – AnirbanMukhopadhyay</i>	

Subject Code		Name of the Subject						L	T	P	C				
		Data Communication						3	0	0	3				
Course Educational Objectives															
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Understand the fundamental concepts and components of data communication and transmission media.														
CO2	Analyze and apply data link layer protocols and error handling techniques for reliable communication.														
CO3	Evaluate network layer protocols, routing algorithms, and IP addressing schemes.														
CO4	Implement transport layer protocols and manage end-to-end communication effectively.														
CO5	Identify and use application layer protocols for various data communication services.														
CO6	Explore advanced topics in data communication and assess their impact on modern network architectures.														
CEO1	To discuss the digital data communication techniques														
CEO2	Gain knowledge on basic concepts of data communication layers, protocols and performance														
CEO3	Understand a few representative protocols and network components														
CEO4	To introduce the functions of different layers from deployed examples														
COURSE EDUCATIONAL OBJECTIVES:															
CO-PO MAPPING															
Cos	PROGRAMMEOUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	1													
CO2	3	1													
CO3	2	2													
CO4	2	3		1											
CO5	2	1													
CO6	2	2													
Avg.	2.3	1.6		1											
SYLLABUS															
UNIT:1 (14 Hours)															
Introduction to Computer Networks, Components, Devices, Topology, OSI model, layers in OSI model, TCP/IP Model, Address types (Physical, Logical, Port, Application-Specific Addresses), Analog and Digital Signal. Transmission Media (Guided Media, Unguided Media). Physical Layer: line encoding, Block coding, Binary data to Digital Signal Conversion, Digital to Analog Signal Conversion, Analog to Analog Signal Conversion, Analogue to Digital Signal conversion. Digital Transmission types (Serial and Parallel), Transmission Modes (Simplex, Half-Duplex and Full-Duplex), Multiplexing (FDM, TDM, WDM, CDM).															
UNIT:2 (10 Hours)															
Switching and Types: Circuit Switched Networks, Datagram Networks, Virtual Circuit Networks.															
Data link layer: Introduction, Error Detection and Correction techniques (Checksum, Cyclic															

Redundancy Check, Hamming Code (1-bit error rectification)), Framing, HDLC Frame Structure and Types, HDLC (NRM, ABM, ARM), Point to Point Protocol, Flow and Error Control, Noiseless Channels protocols (Simplest, stop-and-Wait), Noisy Channels protocols (Stop-and-Wait ARQ, Go-Back-N ARQ, Selective Repeat ARQ),	
UNIT:3	(11 Hours)
Medium Access sub layer: Multiple Access - Random Access Protocols (ALOHA, CSMA (non-persistent, 1-persistent, p-persistent), CSMA/CD, CSMA/CA), Controlled Access Protocols (Reservation, Polling, Token Passing), Channelization techniques (FDMA, TDMA, CDMA). Network Layer: Internetworking, IP Address, IPv4(Header structure, Class types, subnetting), IPv6 (Header structure, Address structure), IPV4 to IPV6 Address Mapping, Address Resolution Protocol, ICMP.	
UNIT:4	(10 Hours)
Transport Layer: User Datagram Protocol, Transmission Control Protocol, Process-To-Process Delivery, Congestion Control, Quality of Service. Application Layer: Domain Name System - Address Resolution, TELNET, E-MAIL (SMTP, POP3), FTP, HTTP, HTTPS, Voice Over IP, Firewall.	
Unit-5	(10 Hours)
Application Layer: Domain Name System - Address Resolution, TELNET, E-MAIL (SMTP, POP3), FTP, HTTP, HTTPS, Voice Over IP, Firewall	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Book: 1. <i>Data Communications and Networking – Behrouz A. Forouzan, Fourth Edition TMH,2006.</i> 2. <i>Computer Networks -- Andrew S Tanenbaum, 4th Edition, Pearson Education.</i>	
Reference Book: 1. <i>An Engineering Approach to Computer Networks-S.Keshav,2nd Edition, Pearson Education</i> 2. <i>Understanding communications and Networks,3rd Edition, W.A.Shay, Cengage Learning.</i> 3. <i>Computer and Communication Networks ,Nader F. Mir, Pearson Education</i> 4. <i>Computer Networking: A Top-Down Approach Featuring the Internet, James F.Kurose, K.W. Ross, 3rd Edition, Pearson Education.</i>	

Subject Code		Name of the Subject								L	T	P	C			
		Data Communication								3	0	0	3			
Course Educational Objectives																
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>																
CO1	Understand the fundamental concepts and components of data communication and transmission media.															
CO2	Analyze and apply data link layer protocols and error handling techniques for reliable communication.															
CO3	Evaluate network layer protocols, routing algorithms, and IP addressing schemes.															
CO4	Implement transport layer protocols and manage end-to-end communication effectively.															
CO5	Identify and use application layer protocols for various data communication services.															
CO6	Explore advanced topics in data communication and assess their impact on modern network architectures.															
CEO1	To discuss the digital data communication techniques															
CEO2	Gain knowledge on basic concepts of data communication layers, protocols and performance															
CEO3	Understand a few representative protocols and network components															
CEO4	To introduce the functions of different layers from deployed examples															
COURSE EDUCATIONAL OBJECTIVES:																
CO-PO MAPPING																
Cos	PROGRAMMEOUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	3	1														
CO2	3	1														
CO3	2	2														
CO4	2	3		1												
CO5	1	1														
CO6	2	2														
Avg.	2.1	1.6		1												
SYLLABUS																
UNIT:1 (14 Hours)																
Introduction to Computer Networks, Components, Devices, Topology, OSI model, layers in OSI model, TCP/IP Model, Address types (Physical, Logical, Port, Application-Specific Addresses), Analog and Digital Signal. Transmission Media (Guided Media, Unguided Media). Physical Layer: line encoding, Block coding, Binary data to Digital Signal Conversion, Digital to Analog Signal Conversion, Analog to Analog Signal Conversion, Analogue to Digital Signal conversion. Digital Transmission types (Serial and Parallel), Transmission Modes (Simplex, Half-Duplex and Full-Duplex), Multiplexing (FDM, TDM, WDM, CDM).																
UNIT:2 (10 Hours)																
Switching and Types: Circuit Switched Networks, Datagram Networks, Virtual Circuit Networks.																
Data link layer: Introduction, Error Detection and Correction techniques (Checksum, Cyclic																

Redundancy Check, Hamming Code (1-bit error rectification)), Framing, HDLC Frame Structure and Types, HDLC (NRM, ABM, ARM), Point to Point Protocol, Flow and Error Control, Noiseless Channels protocols (Simplest, stop-and-Wait), Noisy Channels protocols (Stop-and-Wait ARQ, Go-Back-N ARQ, Selective Repeat ARQ),	
UNIT:3	(11 Hours)
Medium Access sub layer: Multiple Access - Random Access Protocols (ALOHA, CSMA (non-persistent, 1-persistent, p-persistent), CSMA/CD, CSMA/CA), Controlled Access Protocols (Reservation, Polling, Token Passing), Channelization techniques (FDMA, TDMA, CDMA). Network Layer: Internetworking, IP Address, IPv4(Header structure, Class types, subnetting), IPv6 (Header structure, Address structure), IPV4 to IPV6 Address Mapping, Address Resolution Protocol, ICMP.	
UNIT:4	(10 Hours)
Transport Layer: User Datagram Protocol, Transmission Control Protocol, Process-To-Process Delivery, Congestion Control, Quality of Service. Application Layer: Domain Name System - Address Resolution, TELNET, E-MAIL (SMTP, POP3), FTP, HTTP, HTTPS, Voice Over IP, Firewall.	
Unit-5	(10 Hours)
Application Layer: Domain Name System - Address Resolution, TELNET, E-MAIL (SMTP, POP3), FTP, HTTP, HTTPS, Voice Over IP, Firewall	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Book: 3. <i>Data Communications and Networking – Behrouz A. Forouzan, Fourth Edition TMH,2006.</i> 4. <i>Computer Networks -- Andrew S Tanenbaum, 4th Edition, Pearson Education.</i>	
Reference Book: 5. <i>An Engineering Approach to Computer Networks-S.Keshav,2nd Edition, Pearson Education</i> 6. <i>Understanding communications and Networks,3rd Edition, W.A.Shay, Cengage Learning.</i> 7. <i>Computer and Communication Networks ,Nader F. Mir, Pearson Education</i> 8. <i>Computer Networking: A Top-Down Approach Featuring the Internet, James F.Kurose, K.W. Ross, 3rd Edition, Pearson Education.</i>	

Subject Code		Name of the Subject								L	T	P	C		
		Deep Learning								3	0	0	3		
Course Educational Objectives															
CEO1	Understand the concepts of Neural Networks and Deep Learning.														
CEO2	Understand Deep neural network and layered learning approach.														
CEO3	Study and understand CNN and RNN for deep learning.														
CEO4	Learn and understand Auto Encoders and its applications and Understand concept of transfer learning and its applications with keras.														
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Apply basic mathematical concepts in Deep Learning.														
CO2	Work with powerful framework for supervised learning.														
CO3	Deal with Convolution Neural Networks.														
CO4	Analyze various types efficient data encoders, Apply various network models in deep learning.														
CO5	Analyze CNN and their role in implementation														
CO6	Design Enocders and Decoders														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	2	3		3							3	3	3	
CO2	3	3			3							3	3	3	2
CO3	3	3	3		3							3	3	3	3
CO4	3	3		3	3							3	3	3	3
CO5	1	1		3											
CO6	2	1													
Avg.	2.5	2.1	3	3	3							3	3	3	2.6
SYLLABUS															
UNIT:1 (14 Hours)															
Historical trends in deep learning – Machine Learning basics, Learning algorithms – Supervised and Unsupervised Training, Linear Algebra for machine learning, Testing - Cross Validation, Dimensionality Reduction ,Over fitting/Under fitting, Hyper parameters and validation sets, Estimators – Bias – Variance, Loss Function—Regularization, Biological Neuron – Idea of Computational units, McCulloch-Pitts units and Thresholding logic, Linear Perceptron, Perceptron Learning Algorithm, Convergence theorem for Perceptron, Learning Algorithm, Linear Separability, Multilayer perceptron –The first example of network with Keras code, Back propagation.															
UNIT:2 (14 Hours)															
Introduction to Simple DNN, Platform for Deep Learning, Deep Learning Software Libraries, Deep Feed Forward Networks Introduction, Learning XOR, Gradient-based Learning, Various Activation Functions, ReLU, Sigmoid – Error Functions, Architecture Design															
UNIT:3 (10 Hours)															
Differentiation Algorithms, Regularization methods for Deep Learning, Early Stopping, Drop Out, Difficulty of training deep neural networks, Greedy layer wise training, Optimization methods for NeuralNetworks-Adagrad, Adam.															

UNIT:4	(14 Hours)
Convolution Neural Networks Introduction, Convolution Operation, Motivation, Pooling, Normalization, Applications in Computer Vision – ImageNet, Sequence Modelling –VGGNet, LeNet, Recurrent Neural Networks, RNN topologies- Difficulty in Training RNN, Long Short Term Memory, Bidirectional LSTMs, Bidirectional RNNs, Application case study -Handwritten digits recognition using deep learning, LSTM with Keras – sentiment Analysis	
UNIT:5	(14 Hours)
Encoder, Decoder, Auto Encoders Introduction, Auto Encoders, Under Complete Auto Encoders, Regularized Auto Encoders, Stochastic Auto Encoder, Denoising Auto Encoder, Contractive Auto Encoder, Auto Encoder Applications, Dimensionality Reduction and Classification using Auto encoders, Recommendation, Optimization for Deep Learning-Optimizers –RMS prop for RNNs, SGD for CNNs, Application case study – Image dimensionality reduction using encoders ,LSTM with Keras – sentiment Analysis	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Ian Good Fellow, Yoshua Bengio, Aaron Courville,“Deep Learning”, MIT Press,2016. 2. Kevin P. Murphy, “Machine Learning: A Probabilistic Perspective”, MIT Press, 2012.	
Reference Books: 1. <i>Neural Networks: A Systematic Introduction</i>, RaulRojas, 1996. 2. <i>Christopher and M. Bishop , "Pattern Recognition and Machine Learning", Springer Science Business Media, 2006.</i> 3. <i>Jason Brownlee, " Deep Learning with Python", ebook, 2016.</i>	

Subject Code		Subject								L	T	P	C		
		Entrepreneurship Development								3	0	0	3		
Course Educational Objectives															
CEO1	Explore the entrepreneurial mindset and culture that has been developing in companies														
CEO2	Examine the entrepreneurial process involved in both pursuing an entrepreneurial venture within a large company and the creating and managing a new enterprise for implementation of an entrepreneurial venture.														
CEO3	Discuss the dynamics of participating on a business team and the power inherent in a team relative to individual effort														
CEO4	Provide the background and tools necessary to understand and participate in the entrepreneurial process within a large company, in a new venture or as an investor														
Course Outcomes															
CO1	Understand the concepts of entrepreneurship and skills required to be an entrepreneur														
CO2	Identify the Business Opportunities to and evaluate the feasibilities of the Project;														
CO3	Understand the formalities for setting up of a new venture and know the concepts of														
CO4	Evaluate different financial institutions supporting small scale industry.														
CO5	Develop skills of managing Finance, Marketing and HR of a business organization														
CO6	Analyze the impact of emerging trends in entrepreneurship on the global business														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	2				3									
CO2	2	1				0		3			3				
CO3	1	2				0			2		2				
CO4	1	1				0					3				
CO5	1	1				0			3						
CO6	2	1													
AVG	1.33	1.33				0.6		0.6	1		1.6				
SYLLABUS															
Unit-I [10Hrs]															
Concept of Entrepreneurship, Motivation for Economic Development and Entrepreneurial Achievement, Enterprise and Society Why and how to start Business – Entrepreneurial traits and skills, Mind Vrs Money in Commencing New Ventures, Entrepreneurial success and failures, Environmental dynamics and change. Entrepreneurial Process Step by step approach to entrepreneurial start up Decision for Entrepreneurial start up.															
Unit-II [10Hrs]															
Identifying the Business opportunity - Business opportunities in various sectors, formalities for setting up small enterprises in manufacturing and services, Environmental pollution and allied regulatory and non-regulatory clearances for new venture promotion in SME sector. Writing a Business plan, components of a B-Plan, determining Bankability of the project.															
Unit – III [10 Hrs]															
Central / State level Institution promoting SME. Financial Management in small business. Marketing Management, problems & strategies Problems of HRM – Relevant Labour – laws. Sickness in Small Enterprises. Causes and symptoms of sickness – cures of sickness. Govt. policies on revival of sickness and remedial measures.															

Unit – IV

Innovation and Growth Strategies: Innovation and Creativity in Entrepreneurship: Techniques (Design Thinking, Blue Ocean Strategy), Processes, and Best Practices, Strategic Planning for Growth: Market Penetration, Market Development, Product Development, Diversification, Scaling up a Business: Processes, Challenges, and Strategies, Mergers and Acquisitions: Concept, Types, Benefits, and Challenges
Case Studies: Successful Start-ups and Learnings from Failure

Unit – V

Social Entrepreneurship and Sustainable Development- Social Entrepreneurship: Definition, Importance, and Types, Role of Social Enterprises in Community Development, Sustainable Development Goals (SDGs) and Entrepreneurship, Green and Inclusive Entrepreneurship: Concepts, Practices, and Case Studies, Emerging Trends: Digital Entrepreneurship, Women Entrepreneurship, Rural Entrepreneurship

Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs

Text Book:

1. Robert Hisrich, & Michael Peters: Entrepreneurship, TMH, 2009. Dollinger: Entrepreneurship, Pearson, 2009.

Reference Book:

1. Agarwal: Indian Economy, WishwaPrakashan 2009.
2. Dutt&Sundaram: Indian Economy, S.Chand, 2009
3. B D Singh.: Industrial Relations & Labour Laws, Excel, 2009.
4. ArunaKaulgud: Entrepreneurship Management by, Vikas publishing house, 2009.
5. Essential of entrepreneurship and small business management by Thomas W. Zimmerer & Norman M. Searborough, PHI-2009.
6. ND Kapoor: Industrial Law, Sultan Chand & Sons, 2009

VIII Semester [Fourth Year]
B. Tech in Computer Science and Engineering (Data Science) (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	PE	23BCDPE480**	Professional Elective-VI (From Bucket)	3	0	0	3	40	60
2	OE	23BCDOE480**	Open Elective-IV (From Bucket)	3	0	0	3	40	60
3	OE	23BCDOE480**	Open Elective-V (From Bucket)	3	0	0	3	40	60
			TOTAL	9	0	0	9	120	180

PRACTICAL/SESSIONAL									
4	PST	23BCDPW48101	Technical Seminar	0	0	2	1	0	50
5	PST	23BCDPW48102	Major Project Work-II	0	0	6	3	50	50
			TOTAL	0	0	8	4	50	100
			TOTAL	9	0	8	13	170	280

BUCKETS FOR PROFESSIONAL ELECTIVES

Sl. No.	Semester	Professional Elective VI
1	23BCDPE48011	Time Series Analysis & forecasting
2	23BCDPE48012	Nature Inspired computing for Data Science

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Semester	Open Elective IV & V
1	23BCDOE48011	E-Commerce & ERP
2	23BCDOE48012	Blockchain Technology
3	23BCDOE48021	Real Time Systems
4	23BCDOE48022	Embedded Systems
5	23BCDOE48023	Software Testing

Subject Code	Name of the Subject										L	T	P	C	
	Real-Time Systems										3	0	0	3	
Course Educational Objectives															
CEO1: Learn to model the various types of Real Time Systems															
CEO2: Understand the Hardware Requirements.															
CEO3: Acquire Languages on Real Time Systems															
CEO4: Understand and apply different RTS.															
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Explain the fundamentals of Real time systems and its classifications.														
CO2	Understand the concepts of computer control and the suitable computer hardware requirements for real-time applications.														
CO3	Describe the operating system concepts and techniques required for real time systems.														
CO4	Develop the software algorithms using suitable languages to meet Real time applications.														
CO5	Apply suitable methodologies to design and develop Real-Time Systems.														
CO6	Design RTS and Development Methodologies.														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	2	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	3	-	3	-	-	-	-	-	-	-	-	-	-	-
CO5	1	2													
CO6	2	2													
Avg.	2.5	2.5		3											
SYLLABUS															
UNIT:1 (14 Hours)															
Introduction to Real-Time Systems:															
Historical background, Elements of a Computer Control System, RTS- Definition, Classification of Real-time Systems, Time Constraints, Classification of Programs.															
Concepts of Computer Control:															
Introduction, Sequence Control, Loop Control, Supervisory Control, Centralized Computer Control, Hierarchical Systems.															
UNIT:2 (12 Hours)															
Computer Hardware Requirements for Real-Time Applications:															
Computer Hardware Requirements for Real-Time Applications: Introduction, General Purpose Computer, Single Chip Microcomputers and Microcontrollers, Specialized Processors, Process-Related Interfaces, Data Transfer Techniques, Communications, Standard Interface															

UNIT:3 (12 Hours) Languages for Real-Time Applications: Introduction, Syntax Layout and Readability, Declaration and Initialization of Variables and Constants, Cutlass, Modularity and Variables, Compilation of Modular Programs, Data types, Control Structures, Exception Handling, Low-level facilities, Co-routines, Interrupts and Device Handling, Concurrency, Real-Time Support, Overview of Real-Time Language.
UNIT:4 (10 Hours) Operating Systems: Introduction, Real-Time Multi-Tasking OS, Scheduling Strategies, Priority Structures, Task Management, Scheduler and Real-Time Clock Interrupt Handler, Memory Management, Code Sharing, Resource Control, Task Co-Operation and Communication, Mutual Exclusion.
UNIT-5 (10 Hours) Design of RTS- General Introduction: Introduction, Specification Docwnnent, Preliminary Design. Single-Program Approach, Foreground/Background System. RTS Development Methodologies: Introduction, Yow-don Methodology, Ward and Mellor Method, Hately and Pirbhai Method
Teaching Methods: Chalk& Board/ PPT/Video Lectures
Text Books: • <i>Real-Time Computer Control, Stuart Bennet, 2nd Edn. Pearson Education. 2008.</i>
Reference Books: 1. <i>"Real-Time Systems", C.M. Krishna, Kang G Shin, McGraw-Hill International Editions, 1997.</i> 2 <i>Real-Time Systems Design and Analysis, Phillip. A. Laplante, second edition, PHI, 2005.</i> 3. <i>Embedded Systems, Raj Kamal, Tata McGraw Hill, India, third edition, 2005.</i>

Subject Code	Name of the Subject										L	T	P	C	
	Software Testing										3	0	0	3	
Course Educational Objectives															
CEO1: Equip students with a strong foundation in software testing principles, techniques, and tools essential for ensuring software quality.															
CEO2: Develop the ability to design effective test cases, execute testing strategies, and analyze test results to identify software defects.															
CEO3: Enhance problem-solving skills through the application of testing techniques in real-world scenarios, promoting critical thinking and attention to detail															
CEO4: Foster an understanding of the role of software testing in the software development lifecycle, emphasizing collaboration and communication within a development team.															
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Understand the basic concepts, terminology, and importance of software testing within the software development lifecycle.														
CO2	Apply black-box and white-box testing techniques to design effective test cases.														
CO3	Understand the role of test management in a software project and the activities involved in planning and controlling the testing process.														
CO4	Identify the benefits and challenges of test automation in software projects.														
CO5	Explore advanced testing methodologies, including Agile testing, DevOps testing, and continuous integration/continuous testing.														
CO6	Evaluate the emerging trends and challenges in software testing, including testing in AI/ML systems, cloud-based testing, and testing for security vulnerabilities.														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	2	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	3	-	3	-	-	-	-	-	-	-	-	-	-	-
CO5	1	2													
CO6	2	2													
Avg.	2.5	2.5		3				2							
SYLLABUS															
UNIT:1 (14 Hours)															
Software Testing: Introduction, Evolution, Myths & Facts, Goals, Psychology, definition, Model for testing, Effective Vs Exhaustive Software Testing. Software Testing Terminology and Methodology: Software Testing Terminology, Software Testing Life Cycle, Software Testing Methodology.															
Verification and Validation: Verification & Validation Activities, Verification, Verification of Requirements, High level and low level designs, verifying code, Validation.															

UNIT:2	(12 Hours)
Dynamic Testing-Black Box testing techniques: Boundary Value Analysis, Equivalence class Testing, State Table based testing, Decision table based testing, Cause-Effect Graphing based testing, Error guessing. White-Box Testing: need, Logic Coverage criteria, Basis Path testing, Graph matrices, Loop testing, data flow testing, mutation testing.	
UNIT:3	(12 Hours)
Static Testing: Inspections, Structured Walkthroughs, Technical Reviews. Validation activities: Unit testing, Integration Testing, Function testing, system testing, acceptance testing. Regression testing: Progressives Vs regressive testing, Regression test ability, Objectives of regression testing, Regression testing types, Regression testing techniques.	
UNIT:4	(10 Hours)
Efficient Test Suite Management: growing nature of test suite, Minimizing the test suite and its benefits, test suite prioritization, Types of test case prioritization, prioritization techniques, measuring the effectiveness of a prioritized test suite Software Quality Management: Software Quality metrics, SQA models. Debugging: process, techniques, correcting bugs.	
UNIT-5	(10 Hours)
Automation and Testing Tools: need for automation, categorization of testing tools, selection of testing tools, Cost incurred, Guidelines for automated testing, overview of some commercial testing tools such as Win Runner, Load Runner, Jmeter and JUnit. Test Automation using Selenium tool. Testing Object Oriented Software: basics, Object oriented testing Testing Web based Systems: Challenges in testing for web based software, quality aspects, web engineering, testing of web based systems, Testing mobile systems	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1) <i>Software Testing, Principles and Practices</i>, Naresh Chauhan, Oxford 2) <i>Software Testing</i>, Yogesh Singh, CAMBRIDGE	
Reference Books: 1) <i>Foundations of Software testing</i>, Aditya P Mathur, 2ed, Pearson 2) <i>Software testing techniques – Baris Beizer</i>, Dreamtech, second edition. 3) <i>Software Testing, Principles, techniques and Tools</i>, M G Limaye, TMH 4) <i>Effective Methods for Software testing</i>, Willian E Perry, 3ed, Wiley	

Subject Code	Name of the Subject	L	T	P	C										
	E-Commerce & ERP	3	0	0	3										
Course Educational Objectives															
CEO1: Provide students with a comprehensive understanding of the principles and practices of E-Commerce and Enterprise Resource Planning (ERP) systems..															
CEO2: Equip students with the skills to design and implement E-Commerce solutions and ERP systems in various business contexts..															
CEO3: Develop problem-solving abilities by analyzing real-world business scenarios and applying E-Commerce and ERP concepts to optimize business operations.															
CEO4: Prepare students for professional roles in business technology by enhancing their understanding of the integration of E-Commerce platforms with ERP systems															
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Explain the fundamental concepts of E-Commerce, including its types, models, and underlying technologies.														
CO2	Analyze the architecture and functionality of ERP systems and understand how they integrate with various business processes.														
CO3	Design and develop E-Commerce websites and applications using contemporary web technologies and tools.														
CO4	Implement ERP systems in a business environment, configuring modules to suit specific organizational needs.														
CO5	Assess the security, ethical, and legal issues in E-Commerce and ERP, applying best practices to mitigate risks.														
CO6	Evaluate the impact of E-Commerce and ERP on business strategy and operations, recommending improvements for business efficiency and competitiveness														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	2	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	3	-	3	-	-	-	-	-	-	-	-	-	-	-
CO5	1	2													
CO6	2	2													
Avg.	2.5	2.5		3				2							
SYLLABUS															
UNIT:1 (14 Hours)															
Introduction to E-Commerce: Definition, scope, and significance of E-Commerce, E-Commerce business models: B2B, B2C, C2C, C2B, E-Commerce technologies: Internet, intranet, extranet, and web applications, Payment systems in E-Commerce: Credit cards, digital wallets, electronic funds transfer															
UNIT:2 (12 Hours)															
E-Commerce Infrastructure: E-Commerce website development: HTML, CSS, JavaScript, and web hosting, E-Commerce platforms: Shopify, Magento, WooCommerce, E-Commerce databases and data management, User experience (UX) design principles in E-Commerce															

UNIT:3**(12 Hours)**

Introduction to ERP Systems: Definition, scope, and importance of ERP systems, ERP architecture: Client-server, web-based, and cloud-based ERP, Core ERP modules: Finance, HR, manufacturing, supply chain, and CRM, ERP implementation lifecycle: Planning, design, configuration, testing, deployment

UNIT:4**(10 Hours)**

Integration of E-Commerce with ERP: Integration concepts: Data exchange, process synchronization, and unified interfaces, Benefits and challenges of integrating E-Commerce with ERP, Case studies: Successful E-Commerce and ERP integration, Tools and techniques for integration: APIs, middleware, and custom development

UNIT-5**(10 Hours)**

Security, Ethical, and Legal Issues in E-Commerce & ERP: Security concerns in E-Commerce: Data protection, encryption, authentication, ERP security: User roles, access control, and audit trails, Legal issues: Intellectual property, privacy laws, and compliance, Ethical considerations: Data privacy, consumer rights, and digital trust

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Books:

- 1) "Electronic Commerce: A Managerial and Social Networks Perspective"
Authors: Efraim Turban, David King, Jae Kyu Lee, Ting-Peng Liang, Deborrah C. Turban
Edition: 8th Edition (or latest)
- 2) "E-Business and E-Commerce Management: Strategy, Implementation and Practice"
Author: Dave Chaffey Edition: 6th Edition (or latest), Publisher: Pearson

Reference Books:

- 1) "Enterprise Resource Planning: Concepts and Practice" Authors: Vinod Kumar Garg, N. K. Venkitakrishnan Edition: 2nd Edition (or latest), Publisher: PHI Learning



A handwritten signature in blue ink, appearing to read "K. Murali Gopal".

Dr. K.MURALI GOPAL
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